

6.10.10 Preset Output

The preset contents can be output to a printer or as a file.
Outputting the preset contents can facilitate preset management.
For outputting the preset contents, follow the procedures below.

NOTE

- Set the output method from the Destination ID screen (Refer to "6.3.5.3 Destination ID Tab Screen") in the Control Panel screen.

- In the Preset screen, press the [Output] key.

- The preset contents are output to a printer or as a file.



Fig. 6-80

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=====
PRODUCT SET
=====
PRESET NO. 1
===== C 1 =====
PROD NAME :
POTATO CHIPS
PROD CODE :
01
PHOTO NO. : 2
TARGET WT : 90.0 g
UPPER WT : 3.0 g
TOL NEG ERR: 0.0 g
EX. UPPER WT: 0.0 g
SPEED : 70 bpm
DUMP COUNT : 1
INTLK MODE : 1
AV. CONTROL : OFF
SECT. SET : 1
STABLE TIME: 480 ms
METAL HEAD : 0
0 ms

JAW CLOSE TIME:
AMP TIME
RF 1 50 25
RF 2 50 25
RF 3 50 25
RF 4 50 25
RF 5 50 25
RF 6 50 25
RF 7 50 25
RF 8 50 25
RF 9 50 25
RF10 50 25
RF11 50 25
RF12 50 25
RF13 50 25
RF14 50 25
DF 1 50 25
-- DF 1 --
INFEED WT : 500 g
UPPER :+ 20 %
LOWER :- 20 %
500 g
AFD DF UPPER WT :
0 g
AFD DF LOWER WT :
AFD STOP DF LOWER WT: 50 %
WH-DS : 0 MS
IS-WH : 0 MS
-- SECTION 1 --
HEADS : 1- 14
PROD NAME :
PROD CODE :
WH-PH : 190 ms
PH-RF : 130 ms
STAGGER : 0 ms
TARGET WT : 90.0 g
UPPER WT : 3.0 g
TOL NEG ERR: 0.0 g
EX. UPPER WT: 0.0 g
HOP ACTION : 1
FEEDER MULTIPLY: 1
AUTO FD TARGET :
3.8
PHOTO : OFF
SHUTTER : OFF
PRIORITY PARTIC.: 30
AFD :
Manual
AUTO AMP :
MIN. : 0
MAX. : 99
AUTO TIME :
MIN. : 0
MAX. : 99
GOOD EFFCNCY LMT:99.0%

-- OPTIMUM VALUE --
AMP TIME
RF 1 50 25
RF 2 50 25
RF 3 50 25
RF 4 50 25
RF 5 50 25
RF 6 50 25
RF 7 50 25
RF 8 50 25
RF 9 50 25
RF10 50 25
RF11 50 25
RF12 50 25
RF13 50 25
RF14 50 25
DF 1 50 25 (↗)

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Fig. 6-81

6.11 Select Total Pop-up Menu

To display the Select Total pop-up menu, press the [Select Total] pop-up key on the Main Menu. From the displayed keys, select the total data to be displayed.



Fig. 6-82

Table 6-37

No.	Name	Function
1	[Current Total] key	Displays the total and histogram for the period from all total clear to the present.
2	[X-bar Chart] key	Displays the X-bar chart.
3	[Weigher's Transitional Data] key	Displays the operational transition data of the weigher.
4	[Per Head-Cmb. Participation Data] key	Displays the status of each head.
5	[Total Setting] key	Displays the Total Setting screen.
6	[Operation Log] key	Displays the operation log.

NOTE

- The [Total Setting] key is available to "Site Engineer" or higher level personnel.

6.11.1 Current Total Screen

Check the total results in the Current Total screen.

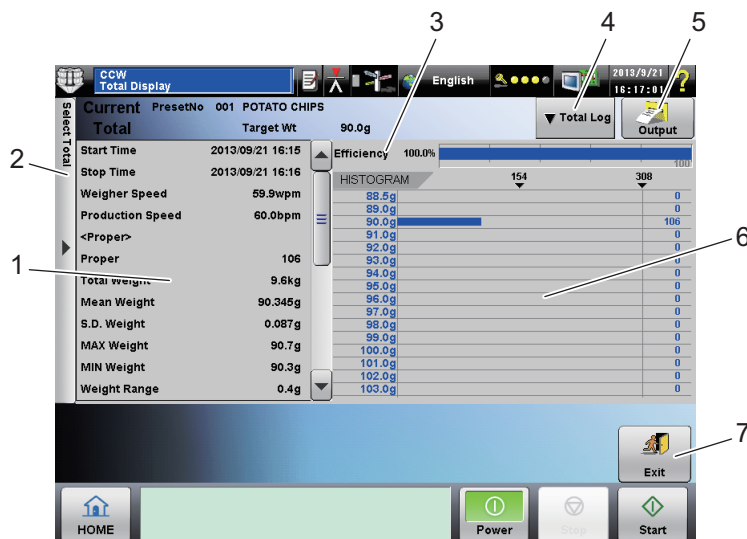


Fig. 6-83

Table 6-38

No.	Name	Function
1	Total data	Displays the production status for the period from the start time of total up to the present.
2	[Select Total] pop-up key	Displays the Select Total pop-up menu.
3	[Effency] display	Displays the efficiency for the period from the start time of total up to the present.
4	[Total Log] drop-down key	Displayed when a total log exists. Selects the total log to be displayed.
5	[Output] key	Outputs total data to a printer or as a file.
6	[HISTOGRAM] display	Displays the histogram.
7	[Exit] key	Returns to the previous screen.

Table 6-39

Item Name	Description
[Start Time]	The time when the first proper dump occurred since combination weighing is started. Display:year/month/day hour:minute
[Stop Time]	The time when combination weighing is stopped or total data output is performed. Display:year/month/day hour:minute

Table 6-39

Item Name	Description
[Weigher Speed]	The frequency of combination weighing in 1 minute every 10 minutes. Unit: wpm
[Production Speed]	The number of productions during operation time. Unit: wpm
[Proper]	The number of proper dumps that occur between the start time and stop time.
[Total Weight]	The total weight of proper dumps that occur between the start time and stop time. Unit: kg ("lb" is used in case of ounce supported)
[Mean Weight]	The mean weight of proper dumps that occur between the start time and stop time. Unit: g ("oz" is used in case of ounce supported)
[S.D. Weight]	The standard deviation weight of proper dumps that occur between the start time and stop time. Unit: g ("oz" is used in case of ounce supported)
[MAX Weight]	The maximum weight of proper dumps that occur between the start time and stop time. Unit: g ("oz" is used in case of ounce supported)
[MIN Weight]	The minimum weight of proper dumps that occur between the start time and stop time. Unit: g ("oz" is used in case of ounce supported)
[Weight Range]	The difference between the maximum weight and minimum weight of proper dumps that occur between the start time and stop time. Unit: g ("oz" is used in case of ounce supported)
[Total Count] * Displayed during count set weighing only.	The total count of proper dumps that occur between the start time and stop time. Unit: pcs
[Mean Count] * Displayed during count set weighing only.	The mean count of proper dumps that occur between the start time and stop time. Unit: pcs
[S.D. Count] * Displayed during count set weighing only.	The standard deviation count of proper dumps that occur between the start time and stop time. Unit: pcs
[MAX Count] * Displayed during count set weighing only.	The maximum count of proper dumps that occur between the start time and stop time. Unit: pcs
[MIN Count] * Displayed during count set weighing only.	The minimum count of proper dumps that occur between the start time and stop time. Unit: pcs
[Count Range] * Displayed during count set weighing only.	The difference between the maximum count and minimum count of proper dumps that occur between the start time and stop time. Unit: pcs
[Mean Piece Weight] * Displayed during count set weighing only.	The mean piece weight of proper dumps that occur between the start time and stop time. Unit: g ("oz" is used in case of ounce supported)

Table 6-39

Item Name	Description
[S.D. Piece Weight] * Displayed during count set weighing only.	The standard deviation piece weight of proper dumps that occur between the start time and stop time. Unit:g ("oz" is used in case of ounce supported)
[MAX Piece Weight] * Displayed during count set weighing only.	The maximum piece weight of proper dumps that occur between the start time and stop time. Unit:g ("oz" is used in case of ounce supported)
[MIN Piece Weight] * Displayed during count set weighing only.	The minimum piece weight of proper dumps that occur between the start time and stop time. Unit:g ("oz" is used in case of ounce supported)
[Parent 1 Total Wt] * Displayed during parent and child weighing only.	The total weight of parent head proper dumps that occur between the start time and stop time. Unit:kg ("lb" is used in case of ounce supported)
[Parent 2 Total Wt] * Displayed during parent and child weighing only.	The total weight of parent head proper dumps that occur between the start time and stop time. Unit:kg ("lb" is used in case of ounce supported)
[Parent 3 Total Wt] * Displayed during parent and child weighing only.	The total weight of parent head proper dumps that occur between the start time and stop time. Unit:kg ("lb" is used in case of ounce supported)
[Parent 4 Total Wt] * Displayed during parent and child weighing only.	The total weight of parent head proper dumps that occur between the start time and stop time. Unit:kg ("lb" is used in case of ounce supported)
[Under]	The number of underweight cases that occur between the start time and stop time.
[Over Weight] <Under/Over>	The number of overweight cases that occur between the start time and stop time as a result of combination.
[Over Scale] <Under/Over>	The number of over-scale cases that occur between the start time and stop time.
[Recheck Error]	The number of recheck errors that occur between the start time and stop time.
[Over Weight] <Over Dump>	The number of overweight dumps that occur between the start time and stop time as a result of combination.
[Over Scale] <Over Dump>	The number of over-scale dumps that occur between the start time and stop time.
[Total Weight] <Over Dump>	The total weight of overweight dumps that occur between the start time and stop time. Unit:kg ("lb" is used in case of ounce supported)
[Metal Detect]	The number of times metal is detected between the start time and stop time. Unit:times When no metal is detected, this item is not displayed.
[Product On Time]	The accumulated time of production between the start time and stop time. Display:hour/minute

Table 6-39

Item Name	Description
[Product Off Time]	The accumulated time of production halt between the start time and stop time. Display:hour/minute
[Low Product Time]	The accumulated time of supply shortage during operation. Display:hour/minute

NOTE

- Parent and child weighing and count set weighing are optional functions (Refer to "8.2 Parent and Child Weighing", "8.3 Count Set Weighing").

6.11.2 X-bar Chart Screen

The X-bar Chart screen displays how weighing result averages change using an X-bar chart.



Fig. 6-84

Table 6-40

No.	Name	Function
1	Upper limit	Set as $+3\Sigma$ (Σ = standard deviation).
2	Mean value	Displays the average of the sum of every weighing result from the total start, per 250 cycles.
3	Lower limit	Set as -3Σ (Σ = standard deviation).

6.11.3 Weigher’s Transitional Data Screen

The Weigher’s Transitional Data screen displays the operational status transition of the device with time. The horizontal axis indicates the time and the vertical axis indicates each value. Up to 4 graphs can be displayed at a time.

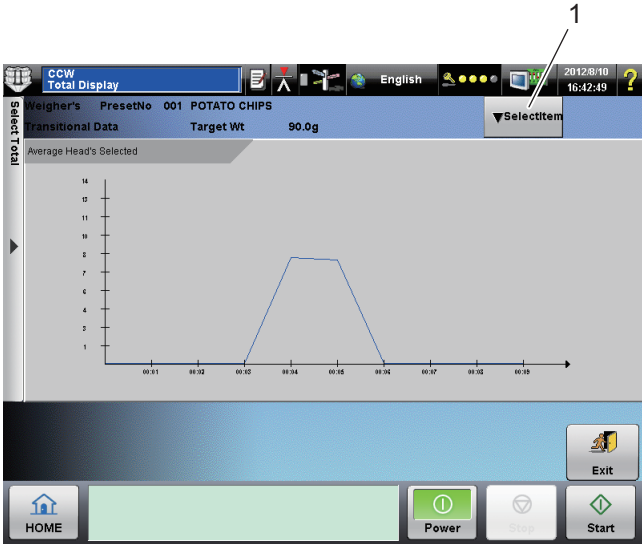


Fig. 6-85

Table 6-41

No.	Name	Function
1	[SelectItem] drop-down key	Selects the items to be displayed from the following (Up to 4 items can be selected). • [Average Head’s Selected] • [Average Stable Heads] • [Average Empty Heads] • [Proper/Over Wt/Under/Over Scale] • [Efficiency] • [Mean] • [Standard Deviation] • [Speed] • [Average Weight per Head] • [Piece Weight Revision] • [Parent Dump Weight]

6.11.4 Per Head-Cmb. Participation Data Screen

The Per Head-Cmb. Participation Data screen displays the operational status of the device for each head. Up to 4 graphs can be displayed at a time.

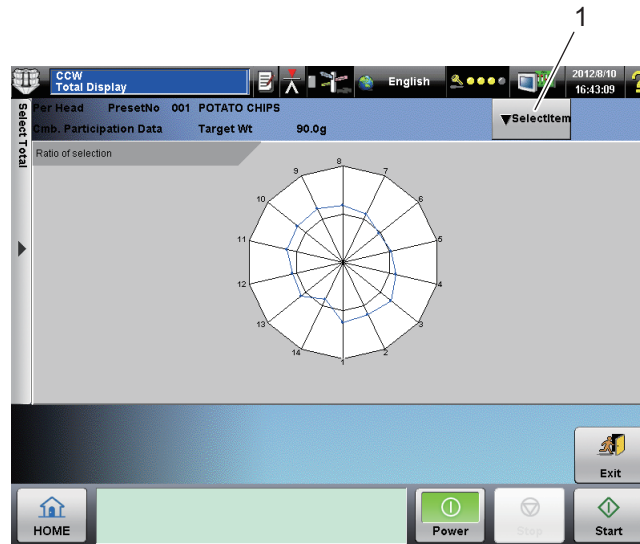


Fig. 6-86

Table 6-42

No.	Name	Function
1	[SelectItem] drop-down key	Selects the items to be displayed from the following (Up to 4 items can be selected). • [Selected Ratio] • [Stable Ratio] • [Empty Ratio] • [Auto Zero Ratio] • [Error Ratio] • [Over Scale Ratio] • [Plus Full Scale Ratio] • [Minus Full Scale Ratio] • [3rd filter Selected Ratio] • [4th filter Selected Ratio] • [Unstable Ratio] • [Average Infeed Product Amount] • [SD Infeed Product Amount]

6.11.5 Total Setting Screen

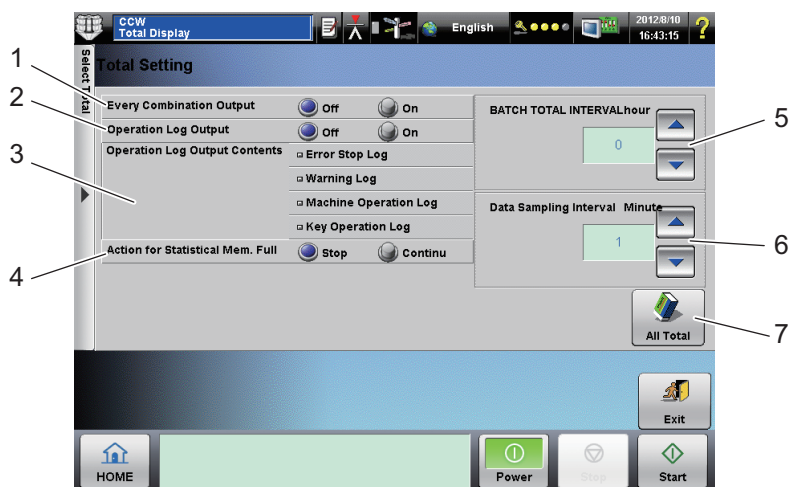


Fig. 6-87

Table 6-43

No.	Name	Function
1	[Every Combination Output] radio button	Selects whether or not to output the combination weight in production. [On]: Outputs the combination weight in production. [Off]: Does not output the combination weight in production.
2	[Operation Log Output] radio button	Selects whether or not to output the operation log. [On]: Outputs the operation log. [Off]: Does not output the operation log.
3	[Operation Log Output Contents]	Selects the contents to be logged, such as the log of commands executed by the device and the errors that occurred during production.
4	[Action for Statistical Mem. Full] radio button	Selects the action to be taken when the memory of total data is full. [Stop]: Stops production. [Continue]: Deletes old data and continues production.
5	[BATCH TOTAL INTERVAL] increase/decrease key	Sets the total data printing interval in increments of 1 hour.
6	[Data Sampling Interval] increase/decrease key	Sets the data sampling interval in increments of 1 minute.
7	[All Total] key	Clears all total data up to the present. Total data cannot be cleared during production.

6.11.6 Operation Log Screen

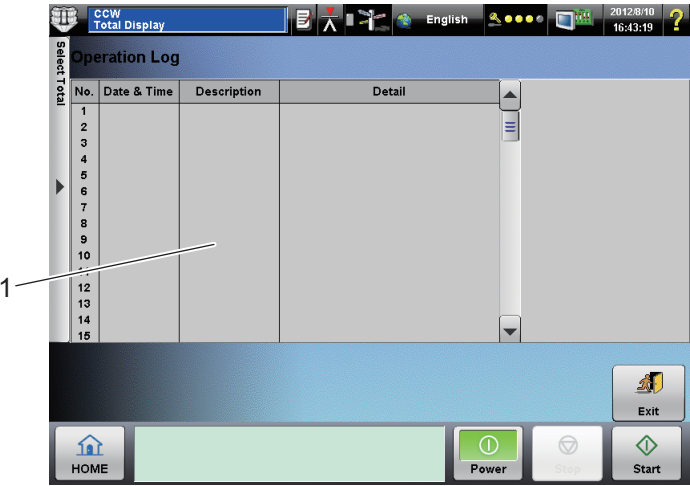


Fig. 6-88

Table 6-44

No.	Name	Function
1	Operation log display area	Displays the contents of the specified log item.

6.12 Feeder Adjustment Screen

The feeder adjustment is the function to adjust the amount of product infeed to the pool hoppers, radial troughs and dispersion tables depending on the operation amplitude and operation time of radial feeder and the operation amplitude, operation time and dispersion weight of dispersion feeder.

The feeder adjustment values depend on the product shape and the target weight. Feeder adjustment is required when the amount of product delivered from the dispersion table to the pool hoppers is too large or too small.

The Feeder Adjustment screen appears when the [Feeder Adjust] tab or the [Feeder Adjustment] index is pressed in the following screens:

- Production screen
- Drain screen
- Full Open Lock Screen
- Preset screen

<Production screen>

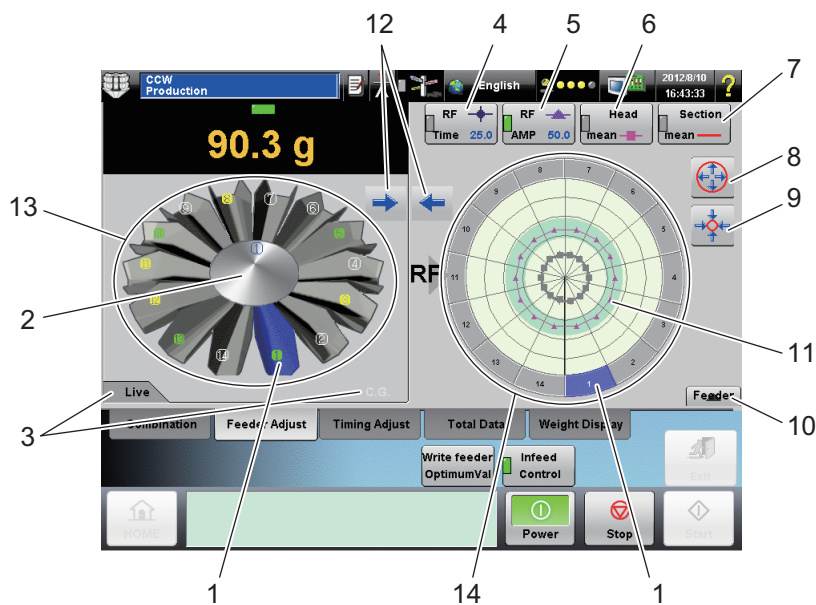


Fig. 6-89

<Drain screen>

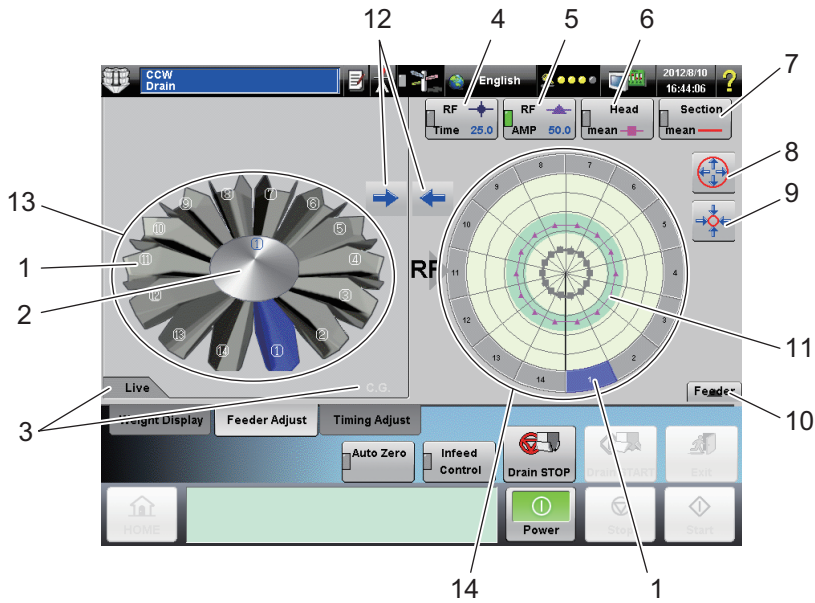


Fig. 6-90

<Full Open Lock screen>

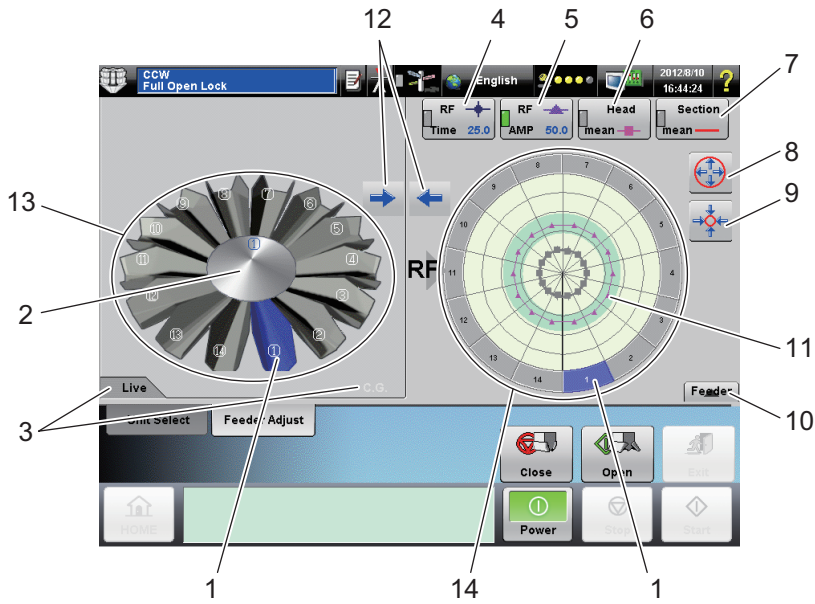


Fig. 6-91

<Preset screen>

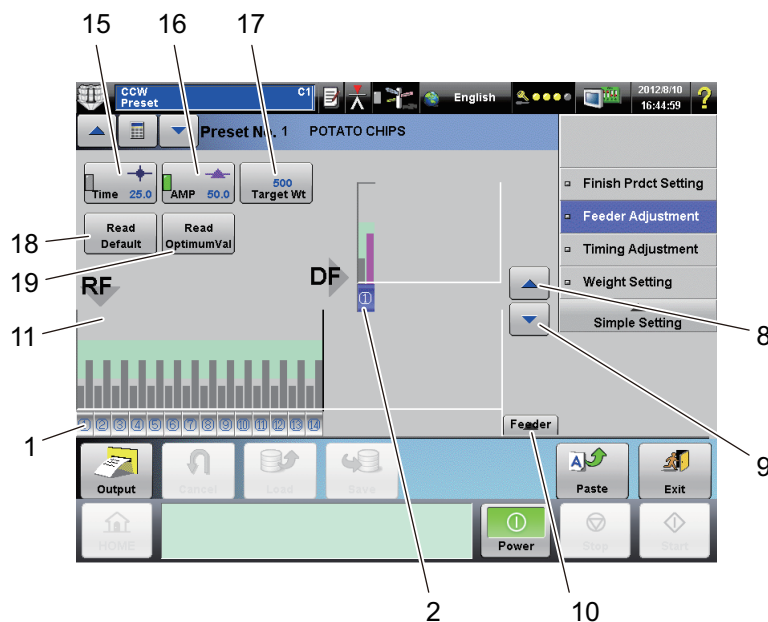


Fig. 6-92

NOTE

- For keys and functions of the Feeder Adjustment index screen during preset, refer to "6.10.3 Feeder Adjustment Index Screen" of the Preset screen.

Table 6-45

No.	Name	Function
1	Each [RF] key	Selects the number for the head to be adjusted. Head selection in the bird's-eye view and the radar chart operates in conjunction with each other.
2	Each [DF] key	Selects the number for the dispersion table to be adjusted. Dispersion table selection in the bird's-eye view and the radar chart operates in conjunction with each other. The feeder adjustment radar chart appears.
3	[Live]/[C.G.] switching tab	Switches the display between the head illustration and the photo taken by the camera.
4	[RF Time] key	ON: Enables adjustment of radial feeder operation time. After turning the key ON, press the [Increase] or [Decrease] key to change the operation time.
5	[RF AMP] key	ON: Enables adjustment of radial feeder amplitude. After turning the key ON, press the [Increase] or [Decrease] key to change the feeder amplitude.

Table 6-45

No.	Name	Function
6	[Head mean] key	ON:Displays the graph of the average infeed product amount for each WH. The values are displayed in pink on the graph while the [Average Infeed Product Amount] key for each WH is selected. If the key is not selected, the values are displayed in grey. The values are indicated by a line connecting each point.
7	[Section mean] key	ON:Displays the graph of the average infeed product amount for each section. The graph is graduated from the center to the periphery to indicate from 0% to 200% of WH capacity, setting the optimum quantity at 100%. The values are displayed in red on the graph while the [Average Infeed Product Amount] key for each section is selected. If the key is not selected, the values are displayed in grey. The values are indicated by a circle.
8	[Increase] key	Increases the feeder value (RF time or RF amplitude) of the selected head.
9	[Decrease] key	Decreases the feeder value (RF time or RF amplitude) of the selected head.
10	[Feeder] pop-up key	Changes display of the Feeder Adjustment screen. Displays the AFD setting menu in pop-up (Refer to "6.12.1 Switching Display in Feeder Adjust Screen").
11	Optimum feeder value range	Displays the optimum feeder value range in pale green while the bar graph and circle graph is displayed. Make adjustment so that the head average and section average is within this range.
12	[Radar chart display switching arrow] key	Switches on/off the radar chart display also on the bird's-eye view.
13	Bird's-eye view	Displays the selection status of the head or dispersion table to be adjusted.
14	Radar chart	Displays the selection status of the head or dispersion table to be adjusted. Displays the feeder time and amplitude of each head or dispersion table.
15	[DF Time] key	ON:Enables adjustment of dispersion feeder operation time. After turning the key ON, press the [Increase] or [Decrease] key to change the operation time.
16	[DF AMP] key	ON:Enables adjustment of dispersion feeder amplitude. After turning the key ON, press the [Increase] or [Decrease] key to change the feeder amplitude.
17	[Dispersion Target Wt] key	Appears if DF is selected. Sets the optimum weight for the dispersion table.
18	[Read Default] key	Calls the feeder default value.
19	[Read OptimumVal] key	Calls the feeder optimum value. Available only when the feeder optimum value is registered in the Production screen (Refer to "6.6 Production screen").

NOTE

- Feeder values changed during production are reflected in the preset.
- Feeder values changed during drain or full open lock are not reflected in the preset.
- For the Feeder Adjustment index screen during preset, refer to "6.10.3 Feeder Adjustment Index Screen".
- The following procedure is explained taking the Feeder Adjust tab screen of the Production screen as an example.

Feeder adjustment procedure

1. In the Feeder Adjust tab screen, press the number for the head to be adjusted (Multiple selection allowed).

► The selected heads are displayed in blue.



Fig. 6-93

2. To adjust the feeder driving time, press the [RF Time] key. To adjust the amplitude, press the [RF AMP] key (Both can be selected).

► The lamp of the pressed key lights up.



Fig. 6-94

3. Press the [Increase] key and [Decrease] key to adjust the time or amplitude.

► The feeder time or amplitude is changed.

► The radar chart changes according to the adjusted value.

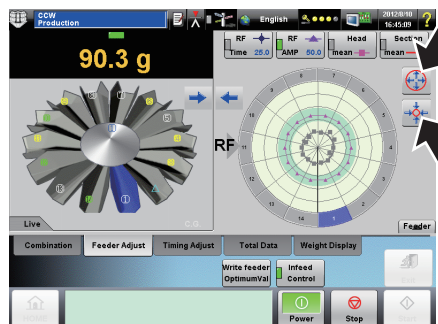


Fig. 6-95

TIP

- When less than 4 heads are selected, the settings will influence the weighing accuracy. Set both of the feeder amplitude and operation time to values about 5 points lower than normal and wait for 1-2 minutes to check the condition.

NOTE

- When feeder control is automatic, the feeder amplitude and feeder operation time settings are used only temporarily and automatic control will be prioritized as time passes.

6.12.1 Switching Display in Feeder Adjust Screen

The Feeder pop-up menu appears when the [Feeder] pop-up key is pressed in the Feeder Adjust tab screen. From the Feeder pop-up menu, the screen can be switched to any of the following displays: production status information, feeder circle graph, feeder bar graph and AFD setting.

- Press the [Feeder] pop-up key.
► The Feeder pop-up menu appears.



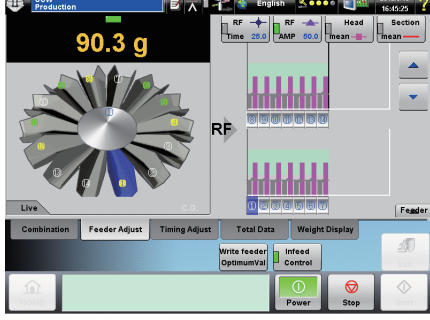
Fig. 6-96

- Select the item to be displayed from the list below.



Fig. 6-97

Table 6-46

No.	Name	Function
1	[Production status information] key	Displays the AFD status and production status.  Fig. 6-98
2	[Circle graph display] key	Displays the feeder value in a circle graph.  Fig. 6-99
3	[Bar graph display] key	Displays the feeder value in a bar graph.  Fig. 6-100
4	[AFD setting] key	Displays the AFD Setting screen (Refer to "6.12.2 AFD Ctroll Mode").  Fig. 6-101

6.12.2 AFD Ctroll Mode

<Function>

This function selects the control method for feeder amplitude, time and dispersion weight. The following 8 modes of control method are available. By referring to the descriptions for each mode mentioned below, select the mode depending on the product characteristics, product infeed status and speed.

Table 6-47 Parameters for Auto Control of Each Feeder

Control Parameter		RF Time	RF AMP	DF Time	DF AMP	DF Weight	Mode
Manual							[Manual]
Auto	Per head		✓			Selectable	[Efficiency Ref. Control]
		✓	✓	✓	✓		[Auto Multi Control]
		✓		✓			[Auto Time Multi Control]
			✓		✓		[Auto AMP Multi Control]
	All heads	✓	✓	✓	✓		[Auto Single Control]
		✓		✓			[Auto Time Single Control]
			✓		✓		[Auto AMP Single Control]

TIP

- Items indicated by blank in the list above are controlled manually.

<For better utilization>

For better utilization of the auto control to realize more stable production, note the following points:

Table 6-48 Notes on Auto Control

Keyword	Points
Preceding infeed status	Stable preceding infeed status is required. In particular, when infeed amount changes drastically, using the auto control may lower the efficiency in some cases. In this case, select the manual mode.

Table 6-48 Notes on Auto Control

Keyword	Points
Dispersion of products	It is important that products are delivered as evenly as possible with no omissions from the dispersion feeder to each radial feeder. Adjust the position of the infeder, feed speed, position of inlet chute, etc., so that products fall onto the center of the dispersion feeder from the infeder and are dispersed as evenly as possible. During feeder parameter setting operations, or when starting production or production operation is unstable, check the flow of the products on the dispersion feeder visually or with a dispersion camera (optional). When there is a problem, take corrective measures such as adjustment of the position of the infeder.
Operating head count	It is required that all heads are ready to operate. If not, the problem has to be solved.
Stable head count	When stable heads required for combination are inadequate due to floor vibration or other causes, the auto control may not work efficiently. In such cases, it is important to have adequate stable heads based on review of speed and filter selection.
Proper cleaning	The auto control function adjusts the feeder capacity in accordance with the changes in flow of products caused by deposits on the table and trough. However, in the case of product flow being stopped by excessive deposits, the auto control cannot solve the problem. It is important to perform proper cleaning in order to maintain high efficiency.
Feeder capacity	The auto control function is based on the assumption that the feed amount of products is increased/decreased in proportion to changes in feeder amplitude. When this condition is not satisfied due to some factors such as products or target weight, the auto control may not operate properly. In this case, select the manual or the auto time control mode.

<Explanation of each control mode>**1. [Manual]**

This mode operates with the feeder amplitude, feeder operation time and dispersion weight values as set. It is not possible to automatically respond to changes in external conditions, such as changes in the amount of supply and the condition of the product on the table and trough which are affected by accumulated deposits. This mode is best suited when making adjustments while checking the status changes in external conditions, when the infeed and product flow is stable. Since it is unlikely that external condition changes will considerably deteriorate the production status, select this mode when the auto control cannot realize stable production.

2. [Efficiency Ref. Control]

The aim of this mode is to perform automatic adjustment corresponding to differences in product characteristics, slight variance in product infeed amount, and changes in infeed amount and flow of products on the table and trough. This mode has the following features (To examine the auto control performance, try this mode first.):

- Controls properly in consideration of weigher status.
- Can automatically adjust amount of products delivered from the dispersion feeder to the radial feeder when the dispersion weight adjustment is selected.

- The control stops during stable status in which the efficiency (Refer to "6.11.1 Current Total Screen") set for the good efficiency judgement value is achieved. However, the control automatically resumes if efficiency lowers due to external condition changes.
- Adjusts feeding capacity of the dispersion feeder to reduce flow increase when the weight of products on the dispersion feeder is larger than the value set for the dispersion weight if the DF adjust for overfeed is selected.

Note the following points in setting up this control mode.

- Set the RF time within the valid time mentioned below in accordance with the product, speed, etc. However, to operate DF continuously, set a value over the valid time below.
[Valid DF (RF) time] = 3600/(Weigher speed)
- Set the DF amplitude so that products can be delivered to all radial troughs, by actually checking the product dispersion status visually or with a dispersion camera (optional).
- Set the AFD range (Refer to "6.12.3 AFD Range Setting") referring to the list below.

Table 6-49 AFD Range

Item Name	Example of Recommended Setting
RF AMP upper limit	Set a larger value based on the average of all RF amplitude values. As the RF amplitude is raised, the corresponding WH weight generally increases accordingly but does not increase over a certain level due to the limit of amount of products delivered from dispersion. Based on this level of amplitude, set a value less than the level.
RF AMP lower limit	Product infeed may stop even if the amplitude is larger than 0 in some cases. Set the lower limit so that such a stop can be prevented.
RF time upper limit, RF time lower limit	Upper limit:99, Lower limit:0
DF AMP upper limit, DF AMP lower limit	Upper limit:99, Lower limit:0
DF time upper limit	Use the figure set for the DF time as an index.
DF time lower limit (Valid only when the DF adjust for overfeed is selected)	Input a value slightly smaller than the value set for the DF time. This mode is effective for maintaining flow volume from DF to RF using products with good dispersion performance, because the DF adjust for overfeed function works in a wide range when the lower limit value is considerably smaller than the specified DF time. For products with poor dispersion performance, set the lower limit to a value as close to the specified DF time as possible.
DF weight upper limit (Valid only when the DF weight adjustment is selected)	It is ideal to set the lower limit to the lowest weight value with which products are stably dispersed from the dispersion feeder to RF and to set the upper limit to the highest weight value with which no unstable factors such as over scale can occur even when the layer thickness is large. However, such conditions are hardly found in actual situations. One of the best ways is to set a range smaller based on the set target weight and then expand it gradually if any problem occurs.
AFD control stop DF weight lower limit [%]	Limit of DF actual weight which stops automatic control. When the DF actual weight is less than the set value, the auto control operation stops. It is considered appropriate to set it to around 50% in normal operation. However, when a low product problem frequently occurs due to unstable infeed, set a smaller value to stop automatic control sooner.

- If layer thickness on the dispersion feeder does not change, even if the DF weight setting is increased, for example, when layer thickness is maintained at a constant level with inlet chute, turn off the DF weight adjustment function. In this case, adjust the DF amplitude/time or inlet chute height so that appropriate amount of products are delivered from the dispersion feeder to the radial feeder.
- This mode is not suitable when the feeder amplitude needs to be fixed to a certain value, for example, in the case that products cannot move unless high feeder amplitude is set. In this case, select the manual or other auto time control mode.

3. [Auto Multi Control], [Auto Time Multi Control], [Auto AMP Multi Control]

Controls each RF independently, as with 2. [Efficiency Ref. Control]. These modes can be selected when automatic and detailed control is desired when infeed and dispersion to each radial feeder are stable.

Especially when the feeder amplitude needs to be fixed to a certain value in cases that products cannot move unless high feeder amplitude is set, select [Auto Time Multi Control]. When the feeder time needs to be fixed during high-speed production or other cases, select [Auto AMP Multi Control].

As an instruction for use, adjust layer thickness on the dispersion feeder by adjusting DF weight or inlet chute height to adjust only the amount of product infeed from the dispersion feeder to the radial feeder, as these modes do not control RF/DF amplitude and time independently. When infeed or dispersion to each radial feeder is unstable or tends to be uneven, the efficiency may be lowered. In this case, select other mode.

4. [Auto Single Control], [Auto Time Single Control], [Auto AMP Single Control]

Unlike other auto control modes, these modes control the whole RF. When 2. [Efficiency Ref. Control] or 3. [Auto Multi Control] is selected, extremely uneven infeed may occur or frequent empty head may occur at a specific head. In such cases, selecting these modes will enable response only to overall status changes generated over a long time due to such causes as influence by deposits. As with 3. [Auto Multi Control], select Auto AMP Single Control when the feeder time needs to be fixed to a certain value during high-speed production or other cases, and select [Auto Time Single Control] when the feeder amplitude needs to be fixed in cases that products cannot move unless high feeder amplitude is set.

As an instruction for use, set the RF amplitude and RF time balance between heads manually in consideration of infeed variance, etc. As with 3. [Auto Multi Control], adjust the amount of product infeed on the dispersion feeder by adjusting DF weight or inlet chute height to adjust only the amount of product infeed from the dispersion feeder to the radial feeder, as these modes do not control RF/DF amplitude and time independently.

<Operation procedure>

1. Press the [Feeder] pop-up key.
 ► The Feeder pop-up menu appears.

**Fig. 6-102**

2. Press the [AFD Setting] key.
 ► The AFD Setting screen appears.

**Fig. 6-103**

3. Select the AFD control mode and press the button.
 ► The selected AFD control mode is confirmed.

**Fig. 6-104**

6.12.3 AFD Range Setting

<Function>

This function sets the limitations in operation when the auto control is selected. When the auto control function is highly responsive to significant external condition changes, the control may not work properly. To prevent such a situation, the control operation can be limited with the AFD range setting. However, bear in mind that the auto control may not work efficiently if the range is too small.

The control values apply to the parameters for auto control of each feeder described in "6.12.2 AFD Ctrll Mode".

1. RF

Table 6-50

Item Name	Function
AMP upper limit, AMP lower limit	RF amplitude is operated within the range between the set upper and lower limits during auto control.
Time upper limit, Time lower limit	RF time is operated within the range between the set upper and lower limits during auto control.

2. DF

Table 6-51

Item Name	Function
AMP	DF amplitude is operated within the range between the set upper and lower limits during auto control.
Time	DF time is operated within the range between the set upper and lower limits during auto control.
Weight [g]	DF weight is operated within the range between the set upper and lower limits during auto control.
AFD control stop DF weight lower limit [%]	Limit of DF actual weight which stops automatic control. When the DF actual weight is less than the set value, the auto control operation stops. e.g.) If the DF target weight is set to 300 g and the AFD control stop DF weight lower limit is set to 60%, the auto control operation stops when the DF actual weight is less than 120 g.

NOTE

- If the setting procedure is described in "6.12.2 AFD Ctrll Mode", follow the descriptions.
- When the RF amplitude and DF amplitude is decreased, product infeed may stop even if the amplitude is larger than 0 in some cases. Set the lower limit so that such a stop can be prevented.
- Note that the DF operates continuously if the DF time is set to a value over the valid time below.
[Valid DF time] = 3600/(Weigher speed)

<Operation procedure>**1) RF AFD range**

1. In the Feeder Adjust screen, press any RF as the head to be adjusted.
2. Press the [Feeder] pop-up key.
 - ▶ The Feeder pop-up menu appears.
3. Press the number for the item to be set from the AFD range setting list.
 - ▶ The Numeric Keypad screen appears.
4. Enter a value within the setting range.
 - ▶ The Numeric Keypad screen disappears, and the entered figures are displayed in the list.

**Fig. 6-105****2) DF AFD range**

1. In the Feeder Adjust screen, press any DF as the head to be adjusted.
2. Press the [Feeder] pop-up key.
 - ▶ The Feeder pop-up menu appears.
3. Press the number for the item to be set from the AFD range setting list.
 - ▶ The Numeric Keypad screen appears.
4. Enter a value within the setting range.
 - ▶ The Numeric Keypad screen disappears, and the entered figures are displayed in the list.

**Fig. 6-106**

6.13 Timing Adjustment Screen

Timing adjustment is the function to adjust the operation timing of each drive unit of the equipment, aiming at efficient product infeed, weighing and discharge.

The Timing Adjustment screen appears when the [Timing Adjust] tab or the [Timing Adjustment] index is pressed in the following screens:

- Production screen
- Drain screen
- Preset screen

NOTE

- The following is explained taking the Timing Adjust tab screen of the Drain screen as an example.

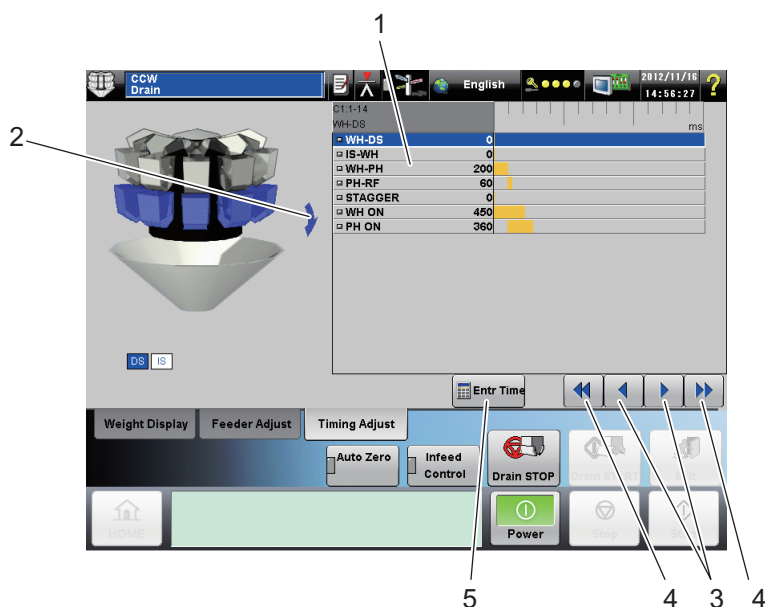


Fig. 6-107

Table 6-52

No.	Name	Function
1	Individual timing adjustment items	Each item can be used as a key to select the item for which timing is to be adjusted.
2	Adjustment direction	Indicates the direction of adjustment between the units displayed in blue.
3	[Increase/Decrease] key (10ms)	Increases/decreases a value in units of 10 ms.
4	[Increase/Decrease] key (100ms)	Increases/decreases a value in units of 100 ms.

Table 6-52

No.	Name	Function
5	[Entr Time] key	Sets the timing value directly from the Numeric Keypad screen.

1. Press each timing adjustment item to select the part to be adjusted.

2. Press  ·  or  ·  to adjust the timing value.
It is also possible to enter the timing value after pressing the [Entr Time] key.

3. To adjust other adjustment items, repeat steps 1 and 2.

To start each function in production, drain or full open lock, you can press the corresponding key at this point to start the operation.

4. To finish the adjustment and return to the Main Menu, press the [Exit] key.

Table 6-53

No.	Item Name	Function
1	WH-DS	Sets the time from the discharge request signal input from the packer up to the discharge completion signal output.
2	IS-WH	Sets the time from the discharge request signal input from the packer up to the WH (weigh hopper) operation start.
3	WH-PH	Sets the time from the WH (weigh hopper) operation start up to the PH (pool hopper) operation start.
4	PH-RF	Sets the time from the PH (pool hopper) operation start up to the RF (feeder) operation start.
5	STAGGER	This item is available when the timing hopper is not used. Used to prevent products from jamming in the packer. Sets the interval for discharging when discharge of the selected head is divided into three times. When not performing staggered discharge, set this item to 0msec.
6	WH ON	Sets the operation time of WH (weigh hopper).
7	PH ON	Sets the operation time of PH (pool hopper).

NOTE

- Timing values changed during production are reflected in the preset.
- Timing values changed during drain are not reflected in the preset.
- Set these parameters with careful attention as they are important parameters.
- Quick-setting enable to automatically optimize timing adjustment parameters.
- Parameters for feeder and filter will also be optimized as well in quick-setting.

6.14 Machine Set Pop-up Menu

The Machine Set pop-up menu is used to configure the settings for the whole device.

To display the Machine Set pop-up menu, press the [Machine Set] pop-up key on the Main Menu. From the menu, select the item to be set and call a desired setting menu.

The items that can be set in the Machine Set pop-up menu are as follows:

- Manual Adjustment (Refer to "6.14.1 Manual Adjustment Screen")
- Self-diagnosis (Refer to "6.14.2 Self-diagnosis Screen")
- Display & Data Manager (Refer to "6.14.3 Display & Data Manager Screen")
- Various Parameter Setting (Refer to "6.14.4 Various Parameter Setting Screen")
- Weigher Setting (Refer to "6.14.5 Weigher Setting Screen")
- Peripheral Equipment Setting (Refer to "6.14.6 Peripheral Equipment Setting Screen")

NOTE

- The [Machine Set] pop-up key is available to "Site Engineer" or higher level personnel.

1. On the Main Menu, press the [Machine Set] pop-up key.

► The Machine Set pop-up menu appears.



Fig. 6-108

2. Press the item to be set.

► The setting screen for the selected item appears.



Fig. 6-109

After completing the setting, press the [Exit] to return to the Main Menu.

6.14.1 Manual Adjustment Screen

To display the Manual Adjustment screen, select [Manual Adjustment] in the Machine Set pop-up menu. The Manual Adjustment screen is used to adjust and display items in the weighing adjustment and the combination calculation screens. Select the screen to be displayed with the tab.

6.14.1.1 Weighing Adjst Tab Screen

The Weighing Adjst tab screen is used for WH manual zero adjustment and span adjustment.

NOTE

- The Weighing Adjst tab screen is available to "Installation" or higher level personnel.

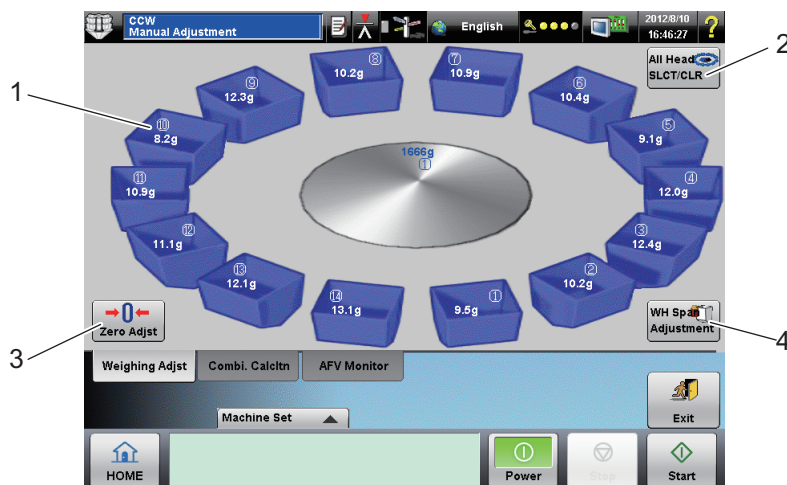


Fig. 6-110

Table 6-54

No.	Name	Function
1	[Head] key	Selects or deselects the head to be adjusted.
2	[All Head SLCT/CLR] key	Selects or deselects all WHs.
3	[Zero Adjst] key	Starts zero adjustment for the selected WH.
4	[WH Span Adjustment] key	Starts span adjustment for the selected WH.

Zero adjustment procedure

1. Press the [ALL Head SLCT/CLR] key.

► All the weigh heads are selected.



Fig. 6-111

2. Press the [Zero Adjst] key.

► The hopper opens/closes, and along with the message "Please wait a moment," the zero adjustment starts.



Fig. 6-112

3. After the zero adjustment is completed, check that the weight display of each weigh head is within $0.0\text{ g} \pm 0.1\text{ g}$.

NOTE

- If the weight display of each head exceeds 0.1 g or falls below -0.1 g , repeat the procedure from step 3.
- The operation details of the zero adjustment procedures (Refer to "6.5 Zero Adjustment screen")
- Next, perform the span adjustment (See the following).



Fig. 6-113

NOTE

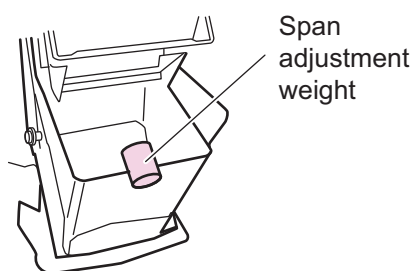
- About zero adjustment
- Zero adjustment is performed while the [Power] key is on.
- Zero adjustment is performed for the weigh head displayed in blue.
- Zero adjustment cannot be executed for the weigh unit and dispersion unit simultaneously.
- When [Multi Dump Initiate] in [Pckr Intrlck Set] is set to [Sync], zero adjustment does not operate without interlock signal from the packer. In this case, interlock the device with the packer.
- Before starting zero adjustment for the weigh unit, products in the weigh hopper are automatically discharged.
- In addition, check the span (See the following).

Span adjustment procedure

1. Place the span adjustment weight on the weigh hopper for which span is to be adjusted.

NOTE

- Normally the span adjustment weight is 200 g. However, it may vary depending on the specifications.

**Fig. 6-114**

2. Select the head to be adjusted.

- The selected heads are displayed in blue. To adjust all heads, press the [All Head SLCT/CLR] key.

**Fig. 6-115**

- Press the [WH Span Adjustment] key.

► The span adjustment starts.



Fig. 6-116

- Make sure that the weight display of the selected head is 200.0 ± 0.1 g.

NOTE

- Make sure that the weight display of the selected head is 200.0 ± 0.1 g. If the weight display reading exceeds 200.1 g or falls below 199.9 g, repeat the procedure from step 2.



Fig. 6-117

- Remove the span adjustment weight from the weigh hopper.

► Make sure that the weight display of the selected head is in the proximity of 0.0 ± 0.1 g.

NOTE

- If the weight display of the selected head exceeds 0.1 g or falls below -0.1 g, repeat the procedure from step 2 in "4.4.7 Zero Adjustment".



Fig. 6-118

6.14.1.2 Combi. Calcltn Tab Screen

To display the Combi. Calcltn tab screen, press the [Combi. Calcltn] tab on the Manual Adjustment screen.

The Combi. Calcltn tab screen displays the results of combination weighings.

NOTE

- The Combi. Calcltn screen is available to "Installation" or higher level personnel.

The functions of the Combi. Calcltn can be selected from the Display/Cal Select pop-up menu.

- Press the [Display/Cal Select] pop-up key.
The Display/Cal Select pop-up menu appears.

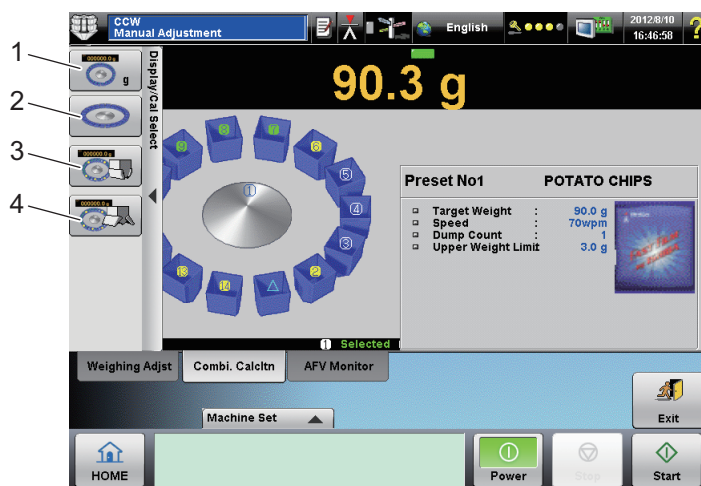


Fig. 6-119

Table 6-55

No.	Name	Function
1	[Combination Result Display] key	Displays the combination result weight.
2	[Combination Status Display with Scale] key	Displayed in the Weight Display screen.
3	[Single Combination Weighing] key	Performs combination weighing only once (Without discharge).
4	[Discharge After Combination Weighing] key	Performs combination weighing only once and discharge after calculation.

6.14.1.3 AFV Monitor Tab Screen

To display the AFV Monitor tab screen, press the [AFV Monitor] tab on the Manual Adjustment screen. The output result of four AFV sensors is indicated in a line graph, from which the weigher vibration status can be checked visually to confirm whether the floor and scaffold where the device is installed are vibrating or not. It is also possible to check the output result of the AFV cell by opening/closing the hoppers or by shaking the weigher.

NOTE

- The AFV Monitor tab screen is available to "Installation" or higher level personnel.

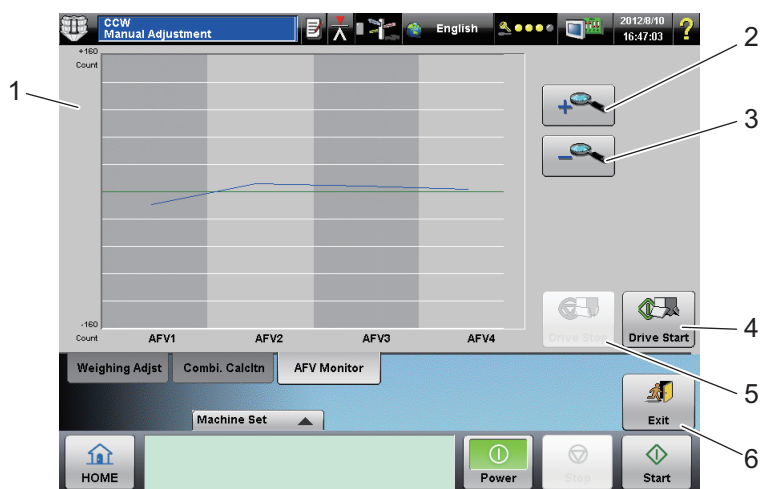


Fig. 6-120

Table 6-56

No.	Name	Function
1	AFV monitor graph	Displays the AFV monitor graph. Display range default value: +160 counts to -160 counts.
2	[Zoom in] key	Each time this key is pressed, the display range is halved. Minimum: +40 counts to -40 counts.
3	[Zoom out] key	Each time this key is pressed, the display range is doubled. Maximum: +1560 counts to -1560 counts.
4	[Drive Start] key	Starts drain.
5	[Drive Stop] key	Stops drain.
6	[Exit] key	Returns to the Main Menu.

6.14.2 Self-diagnosis Screen

To display the Self-diagnosis screen, select Self-diagnosis in the Machine Set pop-up menu.

NOTE

- The self-diagnosis is available to "Installation" or higher level personnel.

The Self-diagnosis screen is used to check operation of the device.

To close the Self-diagnosis screen, press the [Exit] key.

The items that can be diagnosed in the Self-diagnosis screen are as follows:

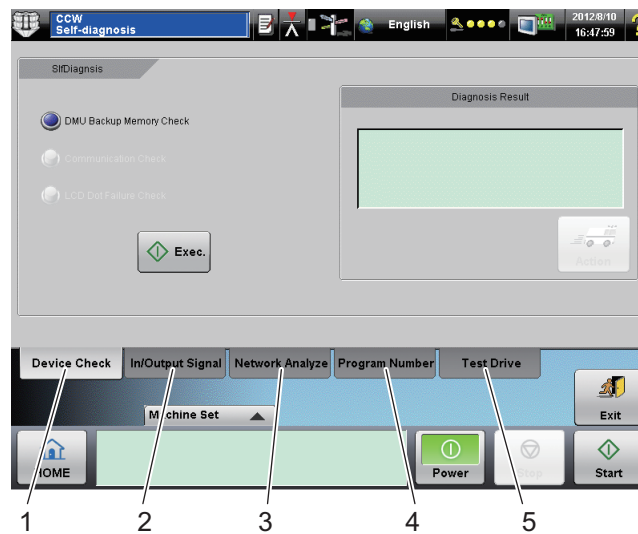


Fig. 6-121

Table 6-57

No.	Name	Function
1	[Device Check] tab	Displays the Device Check tab screen (Refer to "6.14.2.1 Device Check Tab Screen").
2	[In/Output Signal] tab	Displays the In/Output Signal tab screen (Refer to "6.14.2.2 In/Output Signal Tab Screen").
3	[Network Analyze] tab	Displays the Network Analyze tab screen (Refer to "6.14.2.3 Network Analyze Tab Screen").
4	[Program Number] tab	Displays the Program Number tab screen (Refer to "6.14.2.4 Program Number Tab Screen").
5	[Test Drive] tab	Displays the Test Drive tab screen (Refer to "6.14.2.5 Test Drive Tab Screen").

6.14.2.1 Device Check Tab Screen

Performs device check.

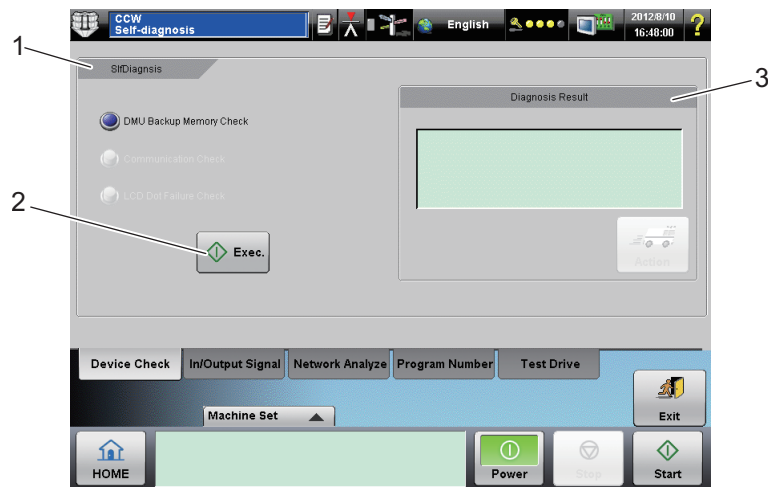


Fig. 6-122

Table 6-58

No.	Name	Function
1	[SelfDiagnosis] radio button	Selects the item to be diagnosed. • [DMU Backup Memory Check]: Performs DMU checksum. If no problem is found as a result of the check, the message “No Problem” is displayed. If there is a problem, the message “DMU Memory Failure. Please initialize memory.” and the problem details are displayed. • [Communication Check]: This is the system in which a network map request is made to WCU and WCU returns network map to DMU. If normal, the message “No Problem” is displayed. If not normal, the message “Some part of Network is disconnected.” is displayed. • [LCD Dot Failure Check]: All light up in white for 3 seconds, then all turn off for 2 seconds. Repeats this process 3 times.
2	[Exec.] key	Starts diagnosis for the selected item.
3	[Diagnosis Result] display	Displays the diagnosis result. If a problem is found, the problem details are displayed in this area and the [Action] key becomes active. If the [Action] key is pressed, the display jumps to the related screen.

6.14.2.2 In/Output Signal Tab Screen

Checks input and output contact signals.

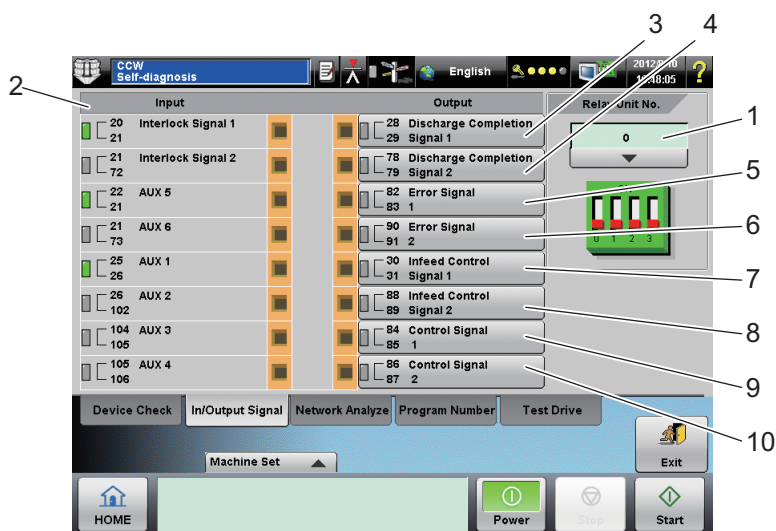


Fig. 6-123

Table 6-59

No.	Name	Function
1	[Relay Unit No.] key	Selects the relay unit number. If a relay unit is selected, the position of the dip switch on the screen changes. The relay unit numbers that can be set are 0-15.
2	Input/output signal display	Displays the relay unit selected in the previous item. On the output side, the output name and the output signal to be tested are selected with the lamp key and the test result is displayed with the lamp. On the input side, the input name and lamp are displayed.
3	[Discharge Completion Signal 1] lamp key	Selects ON/OFF of Discharge Completion Signal 1.
4	[Discharge Completion Signal 2] lamp key	Selects ON/OFF of Discharge Completion Signal 2.
5	[Error Signal 1] lamp key	Selects ON/OFF of Error Signal 1.
6	[Error Signal 2] lamp key	Selects ON/OFF of Error Signal 2.
7	[Infeed Control Signal 1] lamp key	Selects ON/OFF of Infeed Control Signal 1.
8	[Infeed Control Signal 2] lamp key	Selects ON/OFF of Infeed Control Signal 2.
9	[Control Signal 1] lamp key	Selects ON/OFF of Control Signal 1.
10	[Control Signal 2] lamp key	Selects ON/OFF of Control Signal 2.

6.14.2.3 Network Analyze Tab Screen

This screen is used to diagnose the network status of each unit.

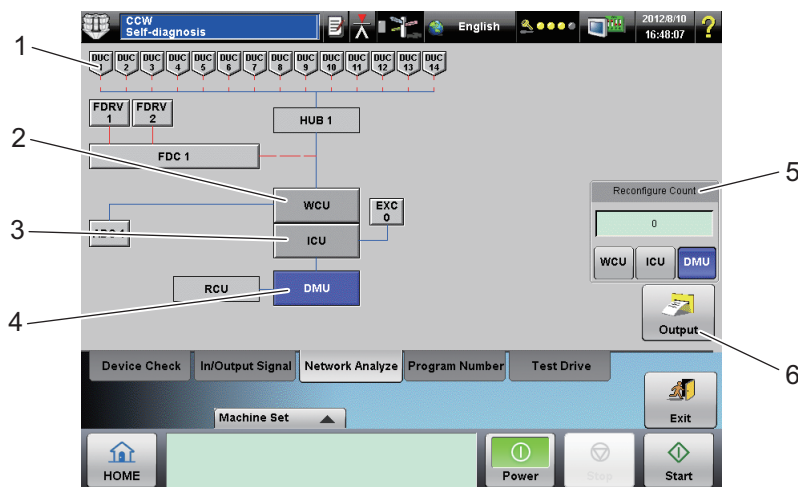


Fig. 6-124

Table 6-60

No.	Name	Function
1	Network map	Displays the network map and a block diagram. Displays each network status of WCU, ICU and DMU. The network status is displayed in the following line colors and line types: Blue line:Normal. Red line:Abnormal (Connected). Red broken line:Abnormal (Disconnected).
2	[WCU] key	Creates the network map around WCU. This includes WCU, ICU, FRDRV, FDC, DUC and ADC.
3	[ICU] key	Creates the network map around ICU. This includes ICU, DMU, WCU and EXC.
4	[DMU] key	Creates the network map around DMU. This includes DMU, ICU and MHIC.
5	Reconfigure Count	Displays the number of times each unit communication is reconfigured.
6	[Output] key	Outputs the status of each unit communication in a two-dimensional graph and the reconfiguration count. *:Network exists. -:No network exists.

6.14.2.4 Program Number Tab Screen

The Program Number screen displays the software information used for this device.

Using this function, program information can be checked without seeing each board mounted on the device. In addition, the information, which is retained until the main power is turned off, can be used to check connection of each unit by comparing it with the network map information.

The Program Number screen displays the program number for each participating network in the order of information received by DMU.

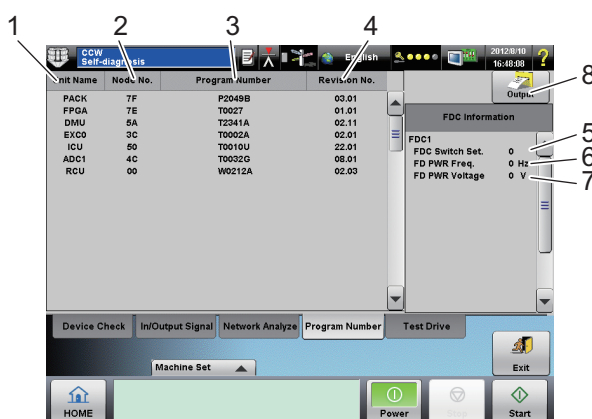


Fig. 6-125

Table 6-61

No.	Name	Function
1	Unit Name display	Displays the unit names of the node numbers listed in the network map table.
2	Node No. display	Displays the node numbers. The numbers are hexadecimal numbers that are identical to the numbers indicated in the network map. For instance, "5" in "5a" corresponds to the vertical axis of the network map and "a" to the horizontal axis.
3	Program Number display	Displays the program numbers. When a program has been upgraded, the version number is attached to the program number.
4	Revision No. display	Displays the revision number of the program in the format of ".*.*".
5	FDC Switch Set.	Displays the FDC switch number for each FDC. Displays the FDC board dip switch settings. The value for the switch is "0" or "4", indicating as follows: 0:FDC not checked, or no phase distributed. 4:Phase distributed.
6	FD PWR Freq.	Displays the feeder frequency value for each FDC.
7	FD PWR Voltage	Displays the feeder voltage value for each FDC.
8	[Output] key	Outputs the program number information to a printer or as a file.

When the received program information is not correct, the incorrect part is indicated with [-----]. The values for the FDC switch, frequency and voltage are all indicated as "0" after the main power is turned on and until the power is turned on.

6.14.2.5 Test Drive Tab Screen

The test drive is used to check feeder operations and hopper opening/closing operations. The test drive should be started after selecting the heads and parts to be used for test drive.

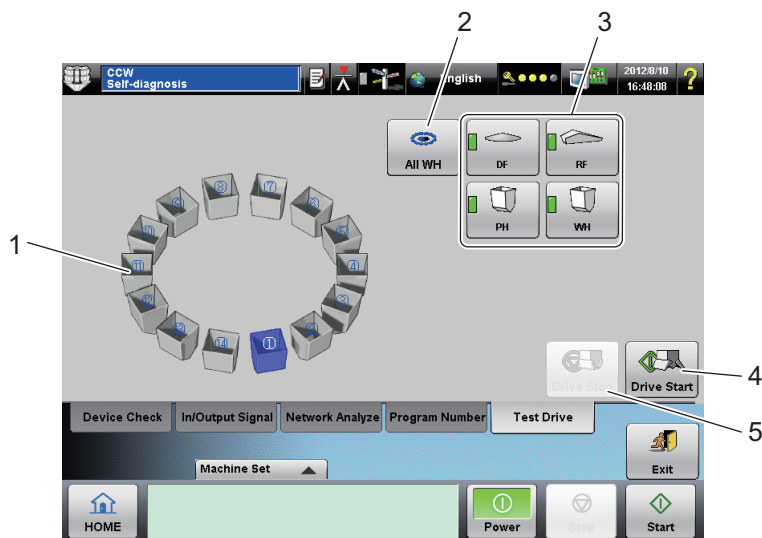


Fig. 6-126

Table 6-62

No.	Name	Function
1	[Head] key	Selects the weigh head.
2	[Slct All WH] key	Selects or deselects all the weigh heads.
3	[Drive Unit] key	Selects the hopper or feeder to be driven. Different keys are displayed depending on the specifications.
4	[Drive Start] key	Starts test drive (Greyed out during drive).
5	[Drive Stop] key	Stops test drive (Greyed out during drive stop).

NOTE

- When the ring shutter, diverting timing hopper or timing hopper is attached as an option, the [RS], [DTH] or [TH] key is added respectively.

Test drive procedure

1. In the Test Drive tab screen, select the hopper or feeder to be driven by pushing the corresponding key.

► The selected hopper or feeder is displayed in blue.



Fig. 6-127

2. Press the [Drive Start] key.

► The test drive starts.

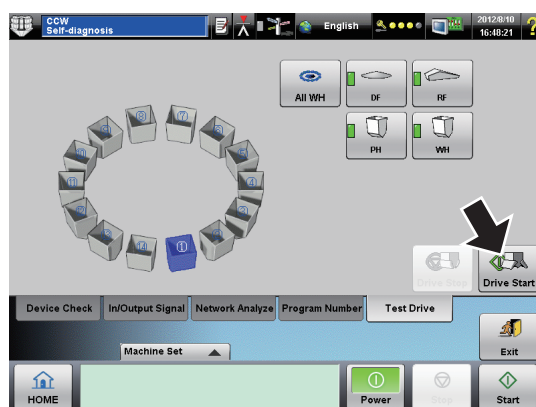


Fig. 6-128

3. To stop the test drive, press the [Drive Stop] key.

► The hopper or feeder drive stops.



Fig. 6-129

6.14.3 Display & Data Manager Screen

To display the Display & Data Manager screen, select "Display & Data Manager" in the Machine Set pop-up menu.

The Display & Data Manager screen is used for layout setting, preset editing and machine configuration editing. Select the screen to be displayed with the tab.

6.14.3.1 Layout Setting Tab Screen

The layout setting screen is used to set the display method in accordance with the location of the remote control unit.



Fig. 6-130

Table 6-63

No.	Name	Function
1	[Head No 1 Location Setting] key	Sets the head to be displayed as head 1 seen from the remote control unit position. Each hopper illustration can be used as a key and is displayed in blue if pressed. With this setting, you can adjust the weigher visually.

6.14.3.2 Preset Manager Tab Screen

Performs preset copy and initialization.

NOTE

- The Preset Manager tab screen is available to "Installation" or higher level personnel.

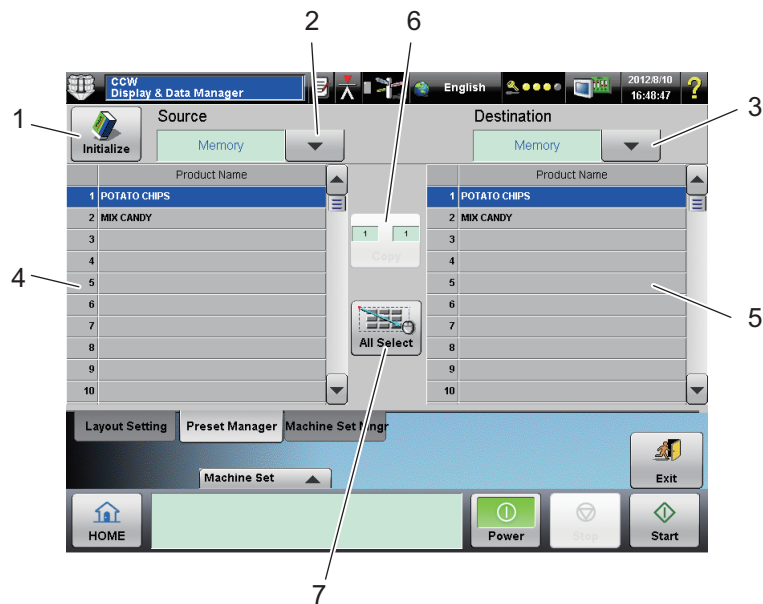


Fig. 6-131

Table 6-64

No.	Name	Function
1	[Initialize] key	Initializes the preset contents of the selected preset number.
2	[Source] drop-down key	Selects "Memory" or "CF/USB" as the copy source.
3	[Destination] drop-down key	Selects "Memory" or "CF/USB" as the copy destination.
4	Copy source preset list	Selects the preset number of the copy source.
5	Copy destination preset list	Selects the preset number of the copy destination.
6	[Copy] key	Copies the preset contents from the source to the destination.
7	[All Select] key	Selects all copy source items.

6.14.3.2.1 Selecting and Copying Preset

This section describes the procedure to select and copy the preset data registered in the memory or USB.

NOTE

- When data is already registered in the specified copy destination, the data is overwritten by the copied data if "Copy" is executed.

- Press the [Source] drop-down key and select the copy source between "Memory" and "CF/USB".



Fig. 6-132

- From the copy source preset list, press the preset number to be used as the copy source.

- ▶ The selected copy source preset item is displayed in blue.
- ▶ On the left of the [Copy] key, the selected copy source preset number is displayed.

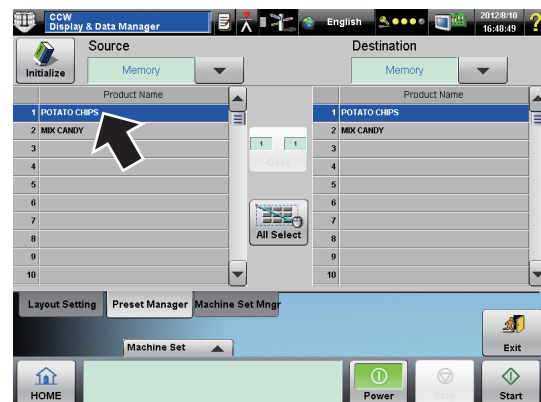


Fig. 6-133

- Press the [Destination] drop-down key and select the copy destination between "Memory" and "CF/USB".

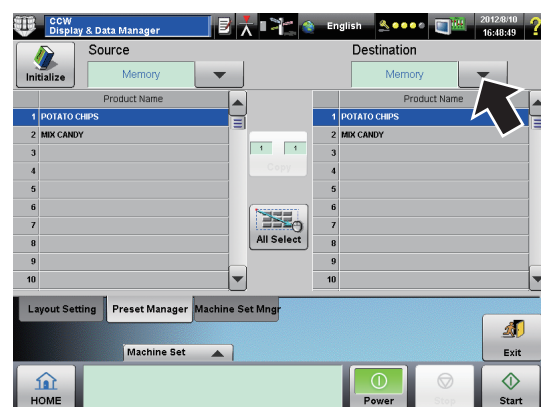


Fig. 6-134

4. From the copy destination preset list, press the preset number to be used as the copy destination.
 - ▶ The selected copy destination preset item is displayed in blue.
 - ▶ On the right of the [Copy] key, the selected copy destination preset number is displayed.



Fig. 6-135

5. Press the [Copy] key.
 - ▶ The confirmation message screen appears.
6. Press the [Yes] key.
 - ▶ The message "Please wait a moment." appears and then copy is performed.



Fig. 6-136

- ▶ The copied preset is displayed in the selected copy destination.
- ▶ The selected preset copy is completed.

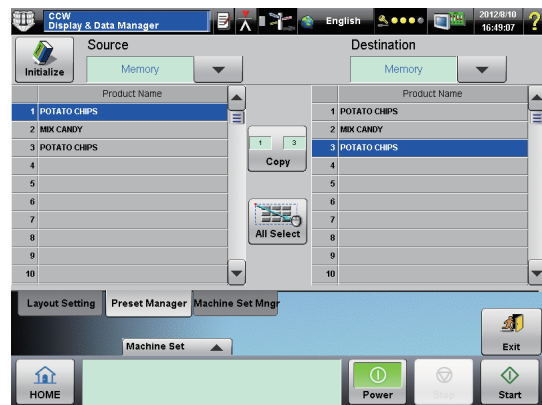


Fig. 6-137

6.14.3.2.2 Copying All Preset

This section describes the procedure to copy all the preset data registered in the memory or USB. In all preset copy, data is copied from memory to USB or from USB to memory.

NOTE

- If all the preset data is copied, original data is overwritten by the copied data and is lost.

- Press the [Source] drop-down key and select the copy source between "Memory" and "CF/USB".



Fig. 6-138

- Press the [All Select] key.
 - ▶ All preset items are displayed in blue.



Fig. 6-139

- Press the [Destination] drop-down key and select the copy destination between [Memory] and [CF/USB].



Fig. 6-140

4. Press the [Copy] key.
 - ▶ The confirmation message screen appears.
5. Press the [Yes] key.
 - ▶ The message "Please wait a moment." appears and then copy is performed.

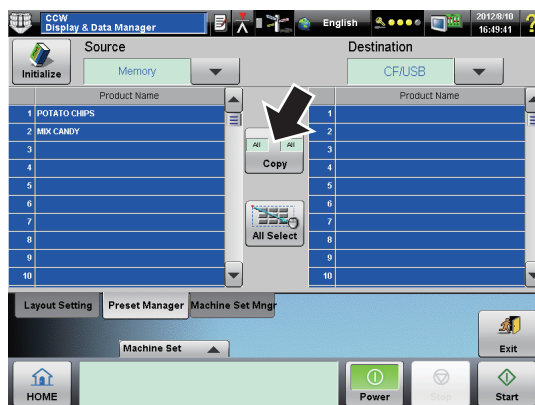


Fig. 6-141

- ▶ The all preset copy is completed.



Fig. 6-142

6.14.3.2.3 Selecting and Initializing Preset

This section describes the procedure to initialize specific preset data.
For initializing preset, the copy source preset list is used.

1. Press the [Source] drop-down key and select the media to be initialized between "Memory" and "CF/USB".

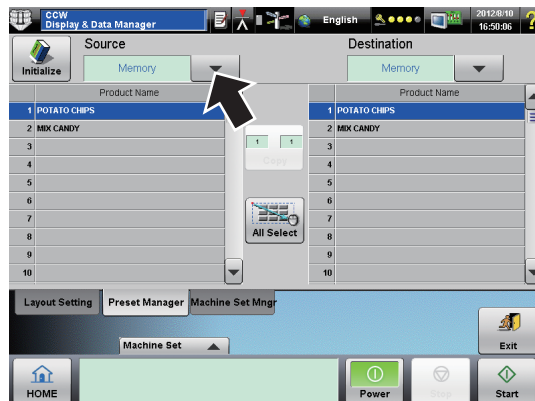


Fig. 6-143

2. From the copy source preset list, press the preset number to be initialized.

► The selected copy source preset item is displayed in blue.

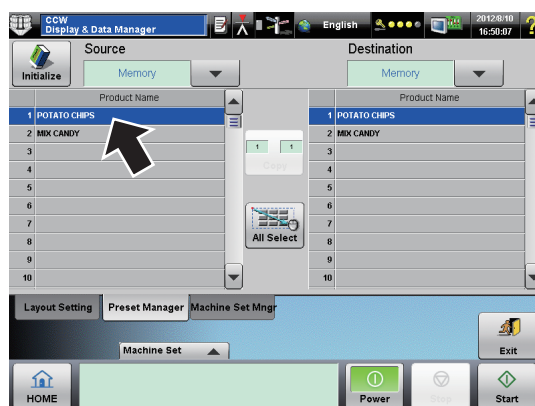


Fig. 6-144

3. Press the [Initialize] key.

► The confirmation message screen appears.
4. Press the [Yes] key.

► The message "Please wait a moment." appears and then initialization is performed.

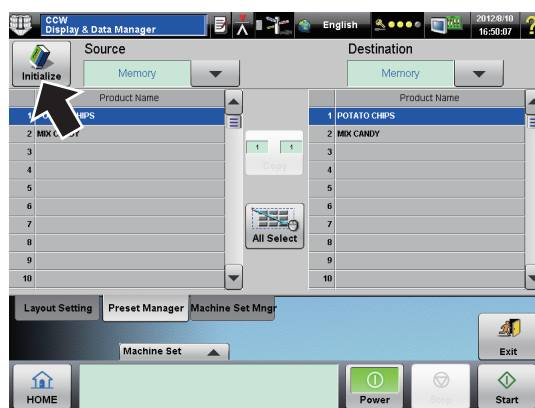


Fig. 6-145

6.14.3.2.4 Initializing All Preset

This section describes the procedure to initialize all preset data.
For initializing all preset, the copy source preset list is used.

1. Press the [Source] drop-down key and select the copy source between "Memory" and "CF/USB".

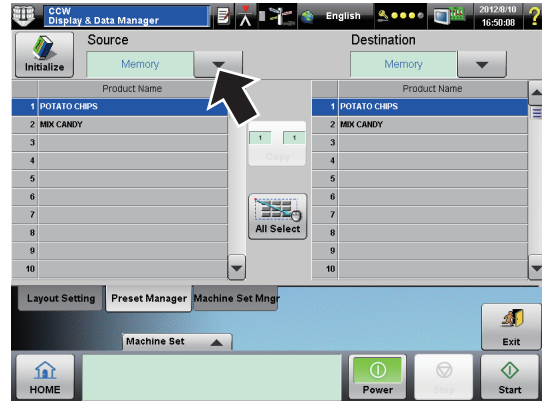


Fig. 6-146

2. Press the [All Select] key.
 - All preset items are displayed in blue.



Fig. 6-147

3. Press the [Initialize] key.
 - The confirmation message screen appears.
4. Press the [Yes] key.
 - The message "Please wait a moment." appears and then initialization is performed.



Fig. 6-148

►The all preset initialization is completed.

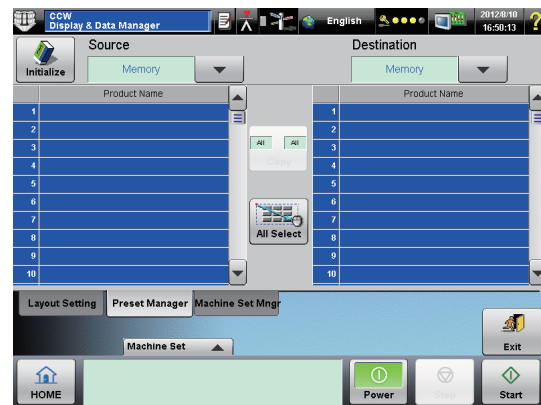


Fig. 6-149

6.14.3.3 Machine Set Mngtr Tab Screen

The Machine Set Mngtr tab screen is used to copy and initialize the machine configuration.

NOTE

- The Machine Set Mngtr tab screen is available to "Installation" or higher level personnel.

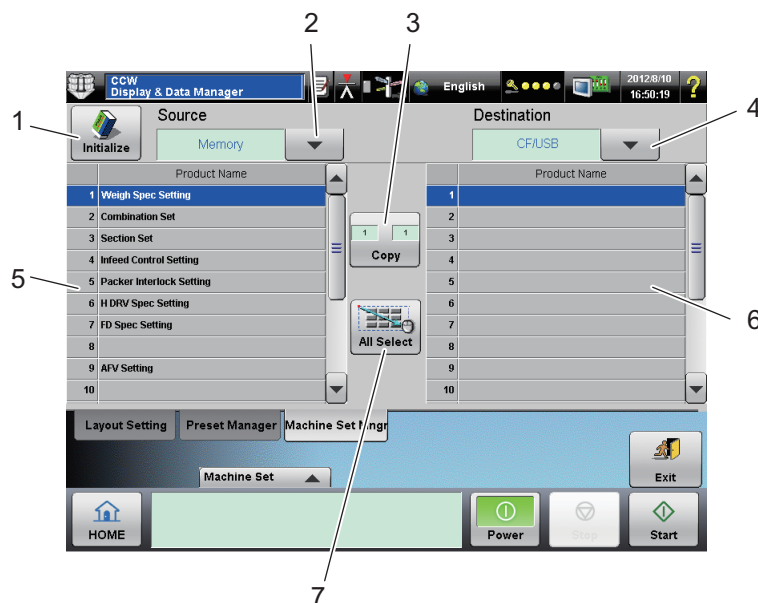


Fig. 6-150

Table 6-65

No.	Name	Function
1	[Initialize] key	Initializes the selected machine configuration.
2	[Source] drop-down key	Selects "Memory" or "CF/USB" as the copy source.
3	[Copy] key	Copies the machine setting items from the source to the destination.

Table 6-65

No.	Name	Function
4	[Destination] drop-down key	Selects "Memory" or "CF/USB" as the copy destination.
5	Copy source machine setting item list	Selects the machine setting item of the copy source. The machine setting items that can be copied are listed below.
6	Copy destination machine setting item list	Displays the machine setting items of the copy destination.
7	[All Select] key	Selects all copy source items.

Table 6-66

No.	Machine Setting Item
1	[Weigh Spec Setting]
2	[Combination Set]
3	[Section Set]
4	[Infeed Control Setting]
5	[Packer Interlock Setting]
6	[H DRV Spec Setting]
7	[FD Spec Setting]
8	(Not used)
9	[AFV Setting]
10	(Not used)
11	[Network setting]
12	[Input Output setting]
13	[Auto timing setting]
14	[Drive Count]
15	[Option setting]
16	[AFD Setting]

NOTE

- Some of the machine setting items cannot be set by the user. Especially in initialization, pay careful attention in selecting items.
- If items that cannot be set by the user are initialized by accident, please contact our service center.

6.14.3.3.1 Selecting and Copying Machine Setting Items

This section describes the procedure to select and copy the machine setting items registered in the memory or USB.

Data is copied from memory to USB or from USB to memory. The machine setting item number is fixed, and it is not possible to copy to a different number.

NOTE

- When an item is already registered, it is overwritten by the copied item.

- Press the [Source] drop-down key and select the copy source between "Memory" and "CF/USB".
It is also possible to press the [Destination] drop-down key and select the copy destination between "Memory" and "CF/USB".



Fig. 6-151

- Select the machine setting item to be copied.
 - ▶ The selected items in both copy source and copy destination are displayed in blue.
 - ▶ In the displays on both sides of the [Copy] key, the selected machine setting item number is displayed.

NOTE

- Only one item can be selected at a time.

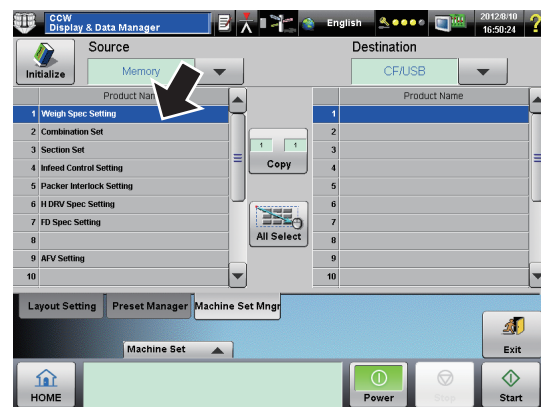


Fig. 6-152

3. Press the [Copy] key.
 ► The confirmation message screen appears.
4. Press the [Yes] key.
 The selected machine setting item is copied.

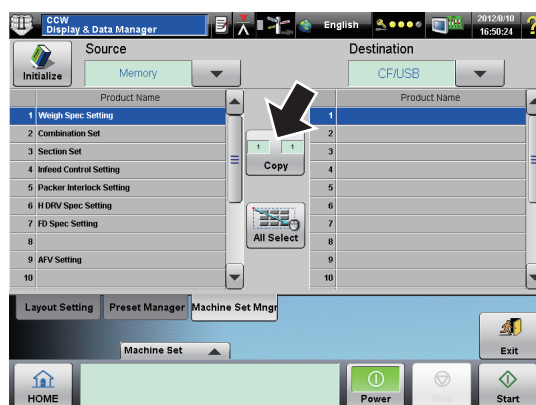


Fig. 6-153

6.14.3.3.2 Copying All Machine Setting Items

This section describes the procedure to copy all the machine setting items registered in the memory or USB.

In all preset copy, data is copied from memory to USB or from USB to memory.

NOTE

- If all the preset data is copied, original data is overwritten by the copied data and is lost.

1. Press the [Source] drop-down key and select the copy source between "Memory" and "CF/USB".
 It is also possible to press the [Destination] drop-down key and select the copy destination between "Memory" and "CF/USB".



Fig. 6-154

2. Press the [All Select] key.

- ▶ All the machine setting items are displayed in blue.
- ▶ In the displays on both sides of the [Copy] key, [All] is displayed.

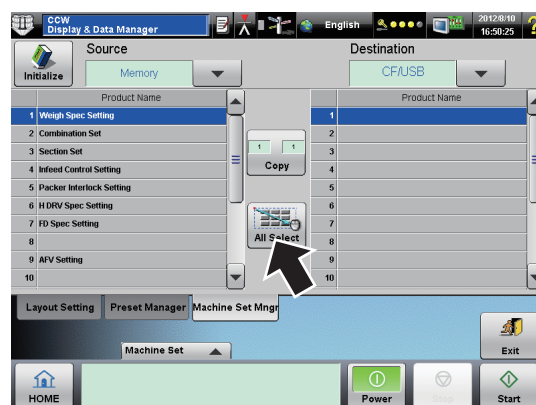


Fig. 6-155

3. Press the [Copy] key.

- ▶ The confirmation message screen appears.

4. Press the [Yes] key.

- ▶ All the machine setting items are copied.

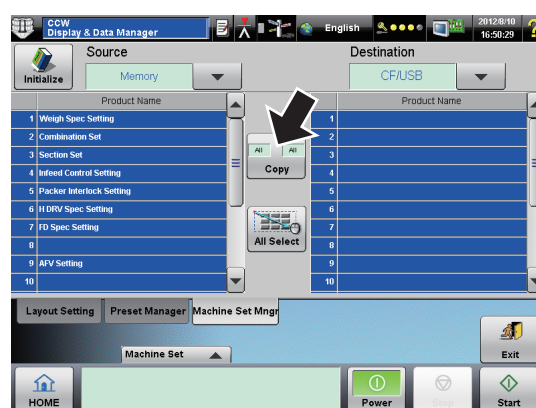


Fig. 6-156

6.14.3.3.3 Selecting and Initializing Machine Setting Items

This section describes the procedure to initialize specific machine setting data.
For initializing machine setting items, the copy source machine setting item list is used.

NOTE

- Once initialization is performed, the data cannot be restored. Before performing initialization, make sure that it will cause no problems.

- Press the [Source] drop-down key and select the media to be initialized between "Memory" and "CF/USB".

NOTE

- If copy destination media is changed, copy source media is also changed. If not noticing this, media that is not to be initialized may be initialized. To prevent accidental deletion of data, do not use the [Destination] drop-down key.



Fig. 6-157

- Select the machine setting item to be initialized.

► The selected item is displayed in blue.

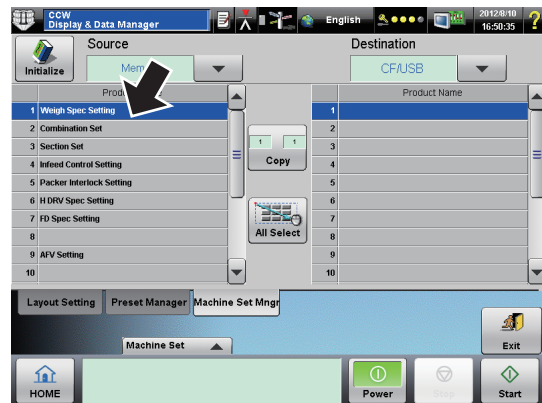


Fig. 6-158

3. Press the [Initialize] key.
 - ▶ The confirmation message screen appears.
4. Press the [Yes] key.
 - ▶ The selected machine setting item is initialized.

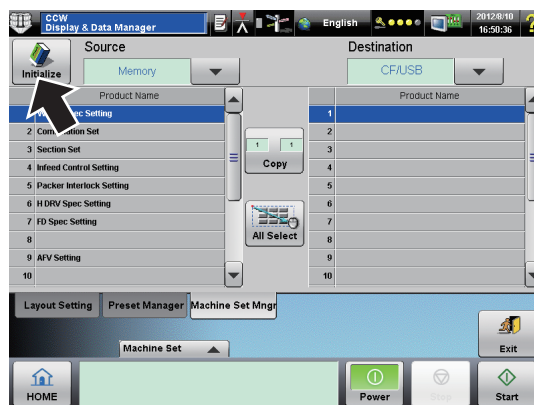


Fig. 6-159

6.14.3.3.4 Initializing All Machine Setting Items

This section describes the procedure to initialize all machine setting data. For initializing all items, the copy source machine setting item list is used.

NOTE

- If initialization is performed, all the machine setting data is lost. Before performing initialization, make sure that it will cause no problems.

1. Press the [Source] drop-down key and select the media to be initialized between "Memory" and "CF/USB".

NOTE

- If copy destination media is changed, copy source media is also changed. If not noticing this, media that is not to be initialized may be initialized. To prevent accidental deletion of data, do not use the [Destination] drop-down key.



Fig. 6-160

2. Press the [All Select] key.

- ▶ All the machine setting items are displayed in blue.



Fig. 6-161

3. Press the [Initialize] key.

- ▶ The confirmation message screen appears.

4. Press the [Yes] key.

- ▶ All the machine setting items are initialized.

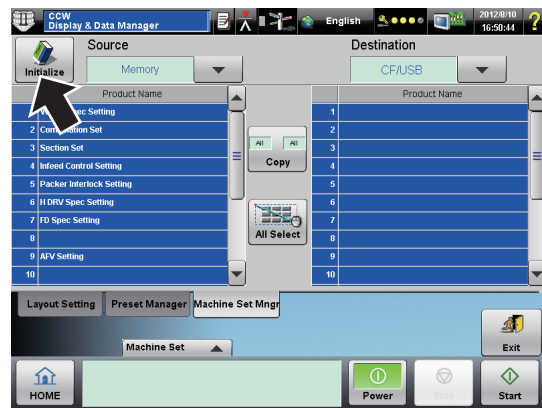


Fig. 6-162

6.14.4 Various Parameter Setting Screen

This screen is used for weigher setting to perform combination weighing inside the weigher.

NOTE

- The Various Parameter Setting screen is available to "Installation" or higher level personnel.

The items that can be set are as follows:

- Weigh Spec Set (Refer to "6.14.4.1 Weigh Spec Set Tab Screen")
- Combination Set (Refer to "6.14.4.2 Combination Set Tab Screen")
- Sectioning Set (Refer to "6.14.4.3 Sectioning Set Tab Screen")

6.14.4.1 Weigh Spec Set Tab Screen

To display the Weigh Spec Set screen, press the [Weigh Spec Set] tab in the Various Parameter Setting menu.



Fig. 6-163

Table 6-67

No.	Name	Function
1	[Range] radio button	Selects the weighing capacity. Selects 400 g or 800 g.
2	[Empty Judgment Weight] key	Sets the weight from which the weigh hopper is judged as empty from the Numeric Keypad screen.
3	[Stable Judgment Weight] key	Sets the weight from which weighing is judged as stable from the Numeric Keypad screen.
4	[Stable Count] key	Sets the number of times to check the stable judgment weight from the Numeric Keypad screen.
5	[Auto Zero Tolerance] key	Sets the tolerance weight to cause no zero error from the Numeric Keypad screen. The auto zero tolerance value can be set in increments of 0.1 g.
6	[Auto Zero Interval] key	Sets the interval of auto zero correction from the Numeric Keypad screen. The auto zero interval is set to "5" in normal operation. If the auto zero interval is set to "0", no auto zero correction is performed.
7	[AFV] radio button	Selects [Off] or [Auto] on the AFV function for each filter number. Sets up each item for Filter 0-Filter 4 (Only for the models with the function).
8	[Output] key	Outputs the set information to a printer or as a file.

6.14.4.2 Combination Set Tab Screen

The Combination Set tab screen is used to set detailed settings related to combination weighing and actions against errors.

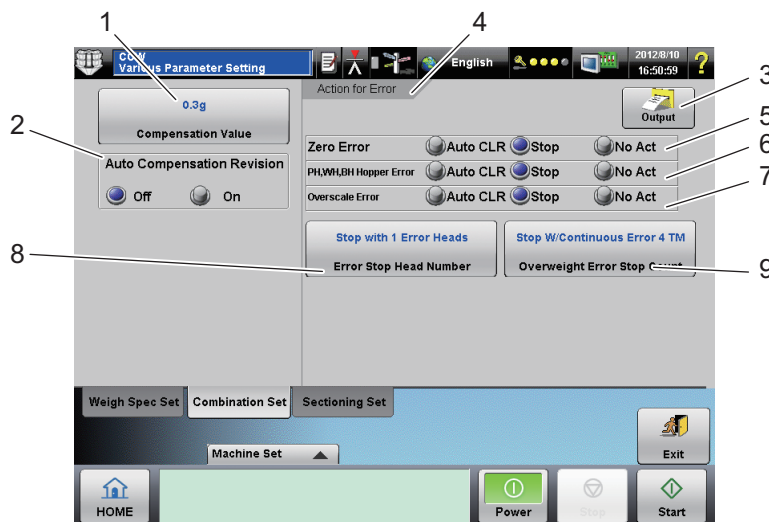


Fig. 6-164

Table 6-68

No.	Name	Function
1	[Compensation Value] key	Sets the compensation value to offset the influence of external vibration from the Numeric Keypad screen. If the weighed value is set to 100.0 g and the compensation value to 0.3 g, combination weighing is performed with the target weight as 100.3 g. The compensation value can be set within the range from 0.0 to 25.5 g, with the minimum setting unit of 0.1 g. The compensation value is set to 0.3 g in normal operation.
2	[Auto Compensation Revision] radio button	Selects whether or not to perform auto compensation revision. The auto compensation revision is the function to minimize reduction in efficiency caused by external vibration as well as to find the optimum compensation value for yield rate and to automatically update it. [On]:Performs auto compensation revision. [Off]:Does not perform auto compensation revision.
3	[Output] key	Outputs the set information to a printer or as a file.
4	[Action for Error]	See the list below.

Table 6-69

No.	Name	Function
5	[Zero Error] radio button	Selects the action for zero error from [Auto CLR], [Stop] and [No Act].
6	[PH,WH,BH Hopper Error] radio button	Selects the action for PH, WH, BH hopper error from [Auto CLR], [Stop] and [No Act].

Table 6-69

No.	Name	Function
7	[Overscale Error] radio button	Selects the action for overscale error from [Auto CLR], [Stop] and [No Act].
8	[Error Stop Head Number] key	Sets the number of permissible head errors from the Numeric Keypad screen. If the number of errors exceeds the set count, production stops. If "0" is set, [AutoDump] is performed.
9	[Overweight Error Stop Count] key	Sets the action for overweight error (i.e. when there are only combination weighing results exceeding the upper limit). If "0" is set, overweight combination discharge is performed regardless of the upper limit. If "1 to 15" is set, production stops when overweight occurs the specified number of times or more consecutively. This is set to "4" in normal operation.

6.14.4.3 Sectioning Set Tab Screen

The Sectioning Set tab screen is used to set the weigh heads to be operated in each section. This setting is normally unnecessary because it is set according to the customer's specification at delivery.

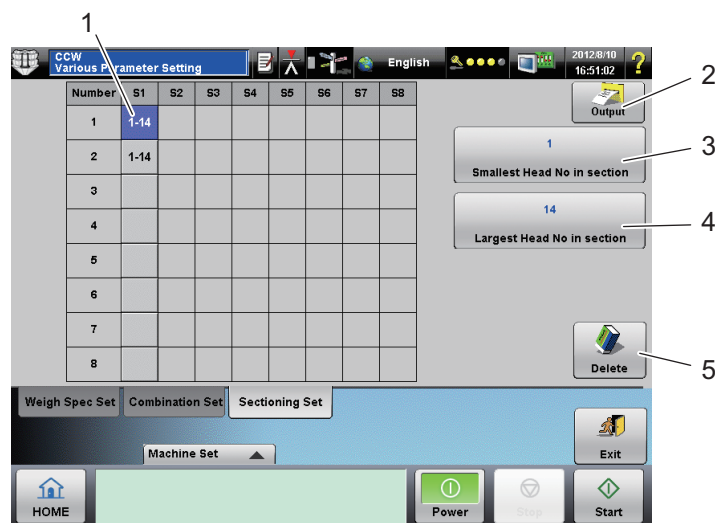


Fig. 6-165

Table 6-70

No.	Name	Function
1	[Parameter Select] key	Selects the section of the desired parameter number.
2	[Output] key	Outputs the set information to a printer or as a file.
3	[Smallest Head No in section] key	Sets the smallest head number of the heads to be operated in the section from the Numeric Keypad screen. Setting range: 1 to max head count.

Table 6-70

No.	Name	Function
4	[Largest Head No in section] key	Sets the largest head number of the heads to be operated in the section from the Numeric Keypad screen. Setting range: 1 to max head count.
5	[Delete] key	Deletes the selected section setting.

6.14.5 Weigher Setting Screen

To display the Weigher Setting screen, press [Weigher Setting] in the Machine Set pop-up menu.

NOTE

- The Weigher Setting screen is available to "Installation" or higher level personnel.

6.14.5.1 Active head Tab Screen

This screen is used to select the hoppers to participate in weighing. This function is used to exclude specific hoppers from participation due to failures or other reasons.



Fig. 6-166

Table 6-71

No.	Name	Function
1	[Head] key	The head of the number pressed participates in weighing. The participating head is displayed in blue.
2	[All Head SLCT/CLR] key	Selects or deselects all heads.

6.14.5.2 AFD Setting Tab Screen

To display the AFD Setting tab screen, press the [AFD Setting] tab in Weigher Setting.
The AFD Setting tab screen is used to configure the AFD setting for the whole weigher.

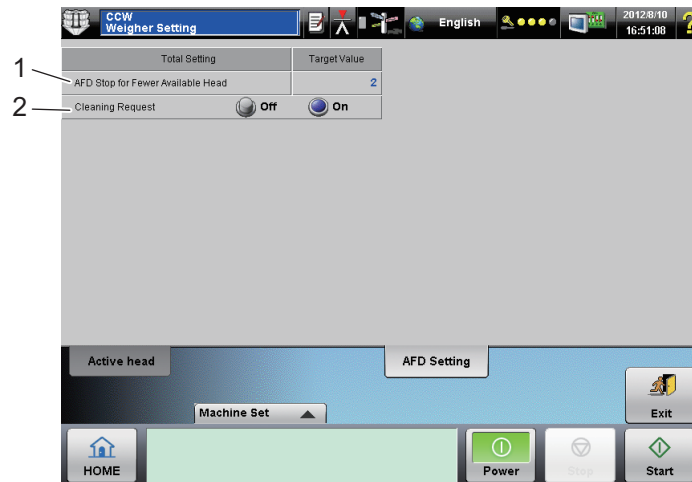


Fig. 6-167

Table 6-72

No.	Name	Function
1	[AFD Stop for Fewer Available Head] key	Sets the number of heads to stop the feeder auto control. The feeder auto control is stopped if the number of heads participating in weighing lowers to the set value or less as a result of non-participation due to a failure, etc. or because weigh hoppers are removed.
2	[Cleaning Request] radio button	If empty head frequently occurs at a certain head, it is judged as defective infeed and a warning is displayed in the Production Status Information screen (Refer to "6.12.1 Switching Display in Feeder Adjust Screen"). [On]:Issues the warning. [Off]:Does not issue the warning.

6.14.6 Peripheral Equipment Setting Screen

To display the Peripheral Equipment Setting screen, select "Peripheral Equipment Setting" in the Machine Set pop-up menu.

The items that can be set in the Peripheral Equipment Setting screen are "Pckr Intrlck Set" and "Infeedr Cntl Set". Select the screen to be displayed with the tab.

NOTE

- The Peripheral Equipment Setting screen is available to "Installation" or higher level personnel.

6.14.6.1 Pckr Intrlck Set Tab Screen

This screen is used to set the configurations for interlocking this device and the packer.

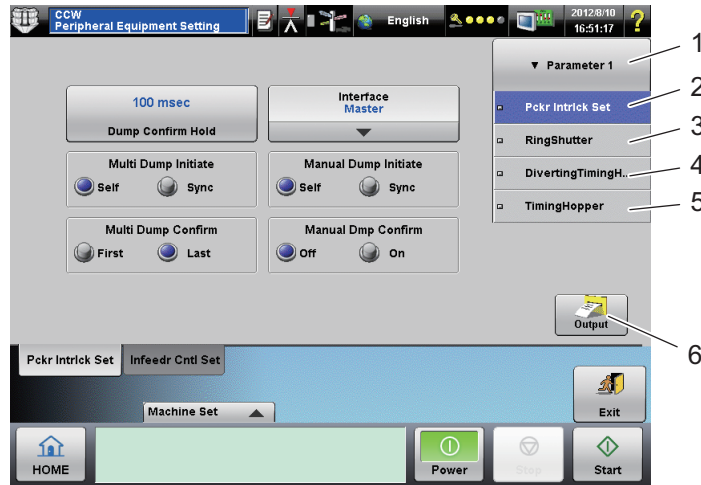


Fig. 6-168

Table 6-73

No.	Name	Function
1	[Parameter No. Select] drop-down key	Selects the parameter number of the packer interlock setting to be set or referred to.
2	[Pckr Intrlck Set] index	Displays the Pckr Intrlck Set index screen (Refer to "6.14.6.1.1 Pckr Intrlck Set Index Screen").
3	[RingShutter] index	Displays the RingShutter screen (Refer to "6.14.6.1.2 RingShutter Index Screen").
4	[DivertingTimingHppr] index	Displays the DivertingTimingHppr screen.
5	[TimingHopper] index	Displays the TimingHopper screen.
6	[Output] key	Outputs the setting contents to a printer or as a file.

TIP

- For diverting timing hopper setting and timing hopper setting, the same procedure is used with the ring shutter setting. Refer to "6.14.6.1.2 RingShutter Index Screen" for setting procedures.

6.14.6.1.1 Pckr Intrck Set Index Screen

This screen is used to set the configurations for interlocking this device and the packer.

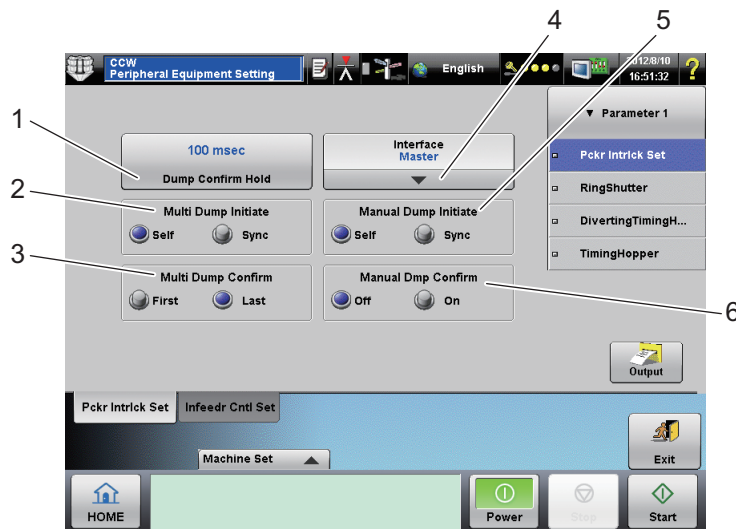


Fig. 6-169

Table 6-74

No.	Name	Function
1	[Dump Confirm Hold] key	Sets the output time of discharge completion signals from the Numeric Keypad screen. Can be set in increments of 10 msec.
2	[Multi Dump Initiate] radio button	Selects the interlock operation signal to use when when the dump count is 2 or more (multiple dump weighing), from [Self] or [Sync]. [Self]:Performs only the first discharge based on the interlock signal from the packer, and the later discharges based on the self-driven timer on this device. [Sync]:Performs all discharges based on the interlock signal from the packer.
3	[Multi Dump Confirm] radio button	Selects the timing to output the discharge completion signal when the dump count is 2 or more (multiple dump weighing), from [First] or [Last]. [First]:Outputs the discharge completion signal after the first discharge. [Last]:Outputs the discharge completion signal after the last discharge.
4	[Interface] drop-down key	Selects the interface for interlocking this device and the packer from [Slave], [Master], [Stroke On Demand] and [Bag On Demand].
5	[Manual Dump Initiate] radio button	Selects the operation of discharge based on the manual interlock signal, from [Self] or [Sync]. [Self]:Performs discharge automatically without input of the interlock signal. [Sync]:Performs discharge based on the interlock signal from the packer.

Table 6-74

No.	Name	Function
6	[Manual Dmp Confirm] radio button	Selects whether or not to output the discharge completion signal during drain, zero adjustment and error discharge. [Off]:Does not output the discharge completion signal during drain, zero adjustment and error discharge. [On]:Outputs the discharge completion signal during drain, zero adjustment and error discharge.

6.14.6.1.2 RingShutter Index Screen

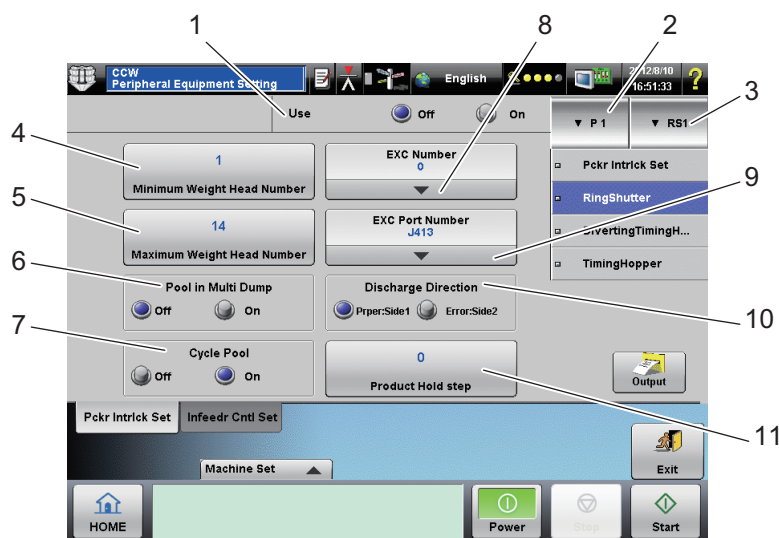


Fig. 6-170

Table 6-75

No.	Name	Function
1	[Use] radio button	Selects whether or not to use the ring shutter.
2	[Parameter Select] drop-down key	Selects the parameter of packer interlock setting.
3	[RS] drop-down key	Selects the ring shutter setting parameter (See Note).
4	[Minimum Weight Head Number] key	Sets the minimum weight head number from the Numeric Keypad screen. Setting range: 1 to max head count.
5	[Maximum Weight Head Number] key	Sets the maximum weight head number from the Numeric Keypad screen. Setting range: 1 to max head count.
6	[Pool in Multi Dump] radio button	Selects whether or not to pool products of amount for one weighing before dump when the dump count is set in the Preset screen. [Off]:Does not pool before dump. [On]:Pools products of amount for one weighing before dump.

Table 6-75

No.	Name	Function
7	[Cycle Pool] radio button	Selects whether or not to pool products of amount for one cycle before dump. [Off]:Does not pool before dump. [On]:Pools products of amount for one weighing before dump.
8	[EXC Number] drop-down key	Selects the EXC number set in the network setting (available to "Maintenance" or higher level personnel).
9	[EXC Port Number] drop-down key	Selects the EXC port number from [J411], [J412], [J413] and [J414].
10	[Discharge Direction] radio button	Selects the discharge direction between [Prper:Side1] and [Error:Side2].
11	[Product Hold step] key	Sets the number of steps seen from WH using the Numeric Keypad screen. Setting range:0-8.

NOTE

The number of RS parameters that can be set is as follows:

- For the minimum weight head number, the maximum weight head number, the pool in multi dump and the cycle pool, 8 patterns can be set for each parameter of packer interlock setting (32 patterns in total).
- For the EXC number, the EXC port number, the discharge direction and the product hold step, a pattern can be set for each parameter of packer interlock setting (4 patterns in total).

The pooling operation depends on the combination of multi dump and cycle as shown below.

Table 6-76

Pool in Multi Dump	Cycle Pool	Operation
[Off]	[Off]	Products are not pooled.
[Off]	[On]	Products of amount for 1 cycle are pooled but not for 1 weighing (multiple dumps).
[On]	[On]	Products of amount for 1 weighing (multiple dumps).
[On]	[Off]	—

TIP

- A cycle refers to a series of process from weighing to dump.
For example, when the dump count is set to "3" for multiple dump weighing, 3 cycles make 1 weighing (corresponding to the target weight).
When the dump count is "1", 1 cycle makes 1 weighing.

6.14.6.1.3 DivertingTimingHppr Index Screen

This screen is used to set the configurations when using the diverting timing hopper.

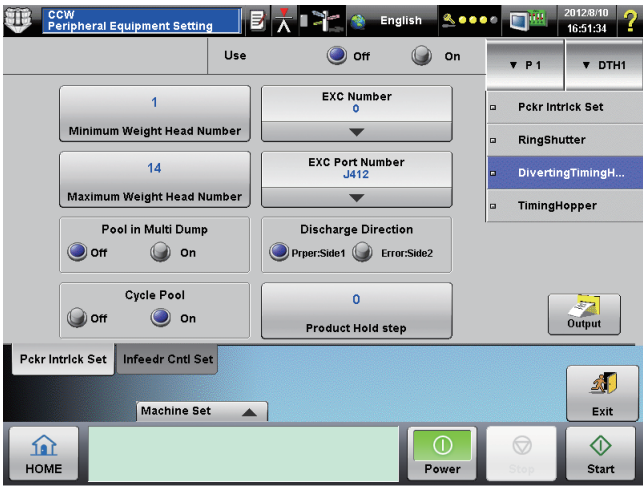


Fig. 6-171

To configure settings, use the ring shutter setting procedure.

6.14.6.1.4 TimingHopper Index Screen

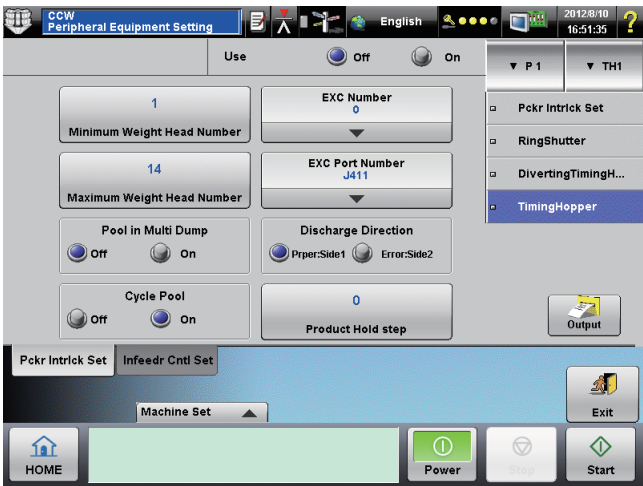


Fig. 6-172

To configure settings, use the ring shutter setting procedure.

6.14.6.2 Infeedr Cntl Set Tab Screen

This screen is used to set the configurations for interlocking this device and the infeedr.

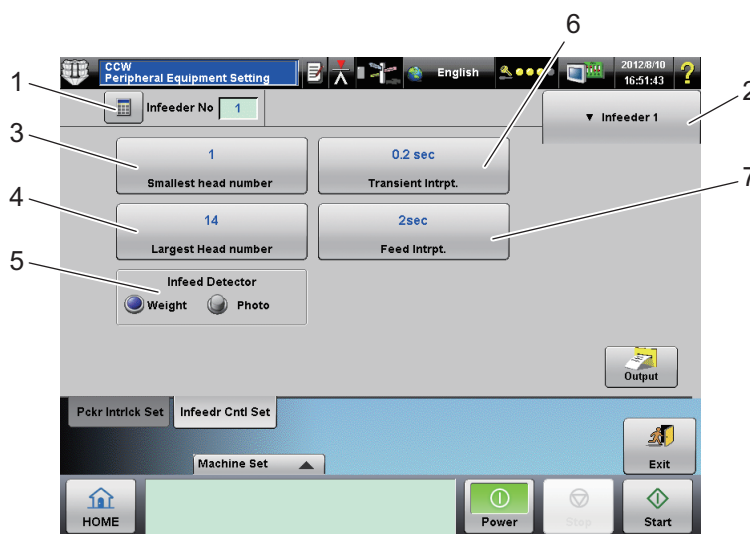


Fig. 6-173

Table 6-77

No.	Name	Function
1	[Infeeder No] key	Sets the number of infeeders to be used for product infeed from the Numeric Keypad screen.
2	[Infeeder No. Select] drop-down key	Selects the infeeder number subject to the following settings.
3	[Smallest head number] key	Sets the smallest head number of the heads in the section to which infeeder 1 or 2 supplies products from the Numeric Keypad screen. Setting range: 1 to max head count.
4	[Largest Head number] key	Sets the largest head number of the heads in the section to which infeeder 1 or 2 supplies products from the Numeric Keypad screen. Setting range: 1 to max head count.
5	[Infeed Detector] radio button	Selects whether to control the products delivered to the Dispersion Table based on weight or volume. [Weight]: Controls the products delivered to the Dispersion Table based on weight. [Photo]: Controls the products delivered to the Dispersion Table based on volume.
6	[Transient Intrpt.] key	Sets the minimum time to judge that the products are loaded on the Dispersion Table from the Numeric Keypad screen. If excess or shortage of products is detected after the set time has passed, the infeeder turns on/off automatically. However, shorter time than the set time is disregarded to prevent the infeeder from turning on/off frequently. Setting range: 0.0 to 25.5 sec. This setting is available only when the infeed detector is in photo mode.

Table 6-77

No.	Name	Function
7	[Feed Intrpt.] key	Sets the time to judge that product infeed from the infeed to the device is interrupted. If product shortage is detected even after the set time has passed, it is judged as interrupted product infeed, and the device stops production, displaying the message "Low Product". When not performing automatic stop, set "0". Setting range:0 to 255 sec.